

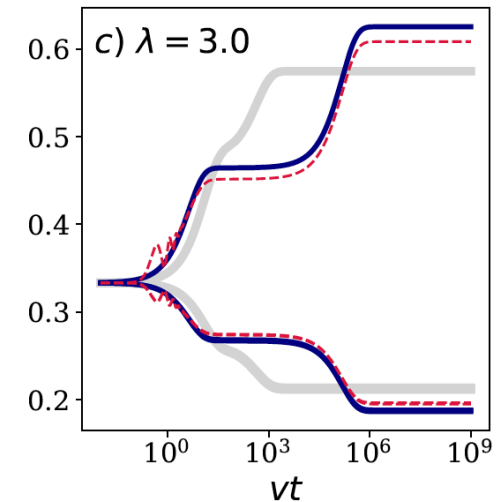
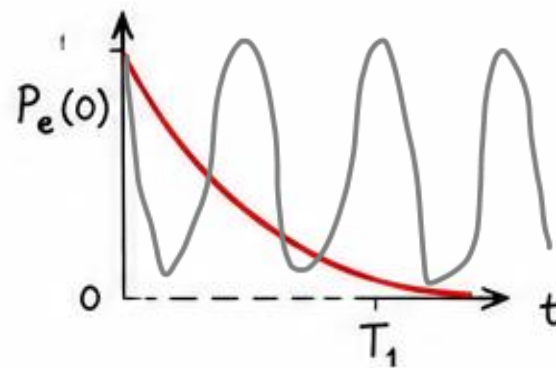
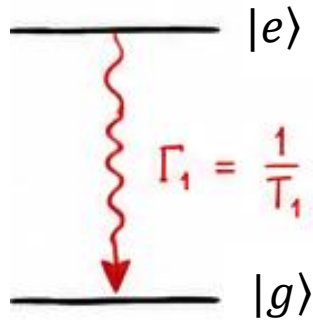
Anomalous quantum relaxation dynamics

Dvira Segal

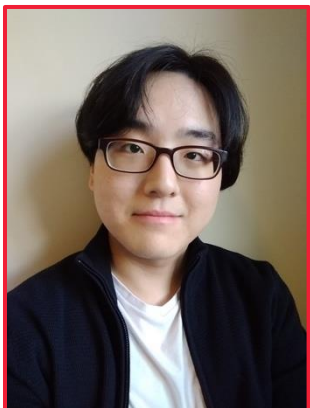
Departments of Chemistry and Physics

Director: CQIQC, QAT

University of Toronto



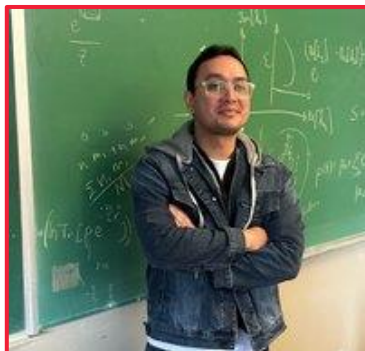
UNIV
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(UG→med school)



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Prositto (Grad)



Madeline Forbes
(UG UBC)



Matthew Pocrnic (GS)



Prof. Nir Bar-Gill
(HUJI)



Roi Nevo
(grad student)

1. Method development (Open Q Systems - Many Body physics)

Goal: Understand effects emerging from strong coupling, finite bath, nonthermal bath, memory effects, correlated bath,...

Projects: Develop new analytical/semi-analytical/computational methods.

Tools: (smart) perturbative methods, Tensor networks, path integrals, stochastic approximations



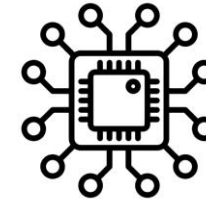
Some projects are done in collaboration with experimentalists on superconducting qubits, NV centers, atomic scale junctions

2. Quantum Computing: noise and algorithms

Goal:

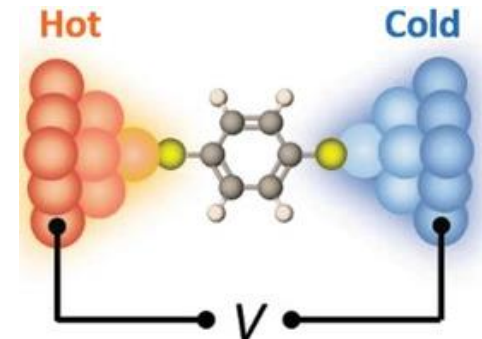
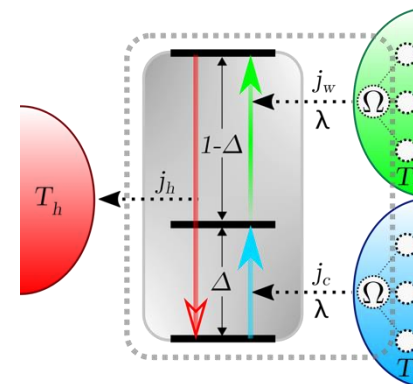
- * quantum algorithms for open quantum systems
- * Error mitigation approaches
- * Resource estimation: Energy considerations in quantum computing
- * quantum simulations on current devices

Collaborators: Quantum Software Consortium, NRC Waterloo



3. Thermal transport, noise

- Energy-precision tradeoffs in classical and quantum systems
- Quantum machines
- Phononic thermal transport



Outline: Today's talk

○ Introduction

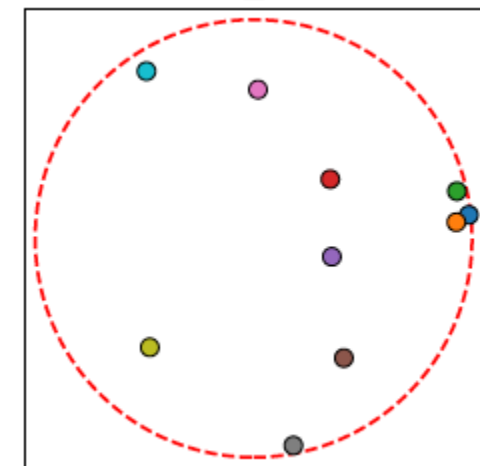
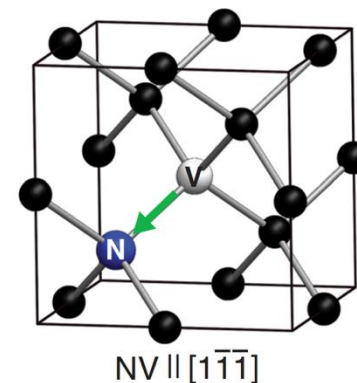
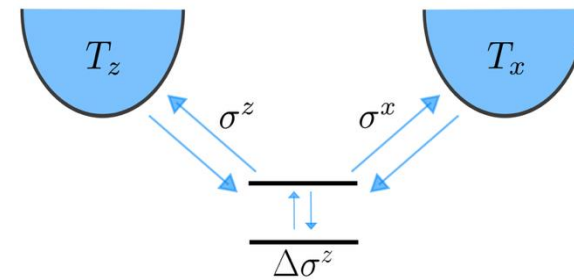
- T1 and T2 timescale and boring quantum dynamics

○ Strong coupling: -Nonadditive effects

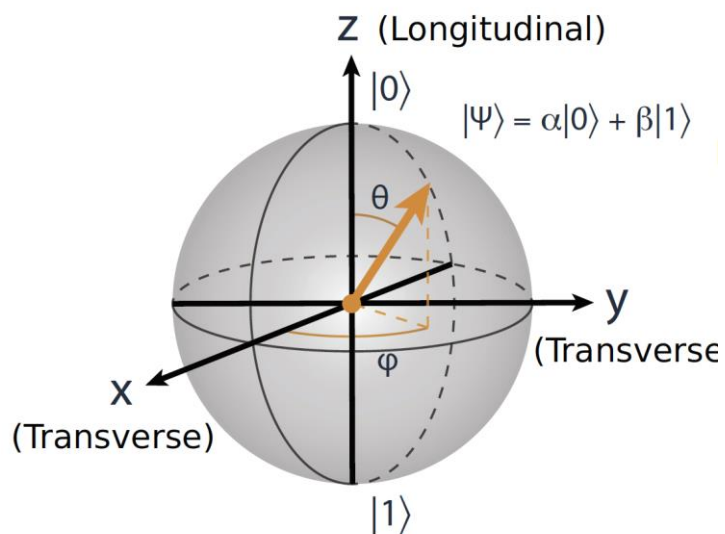
-Slow down - separation of relaxation timescale

○ Geometric heterogeneity: disorder with dephasing leading to long lived dynamics

○ Summary



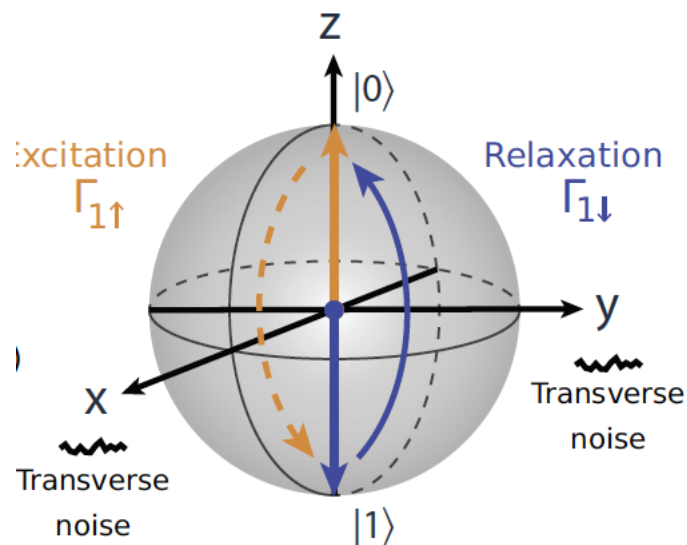
(a) Bloch sphere



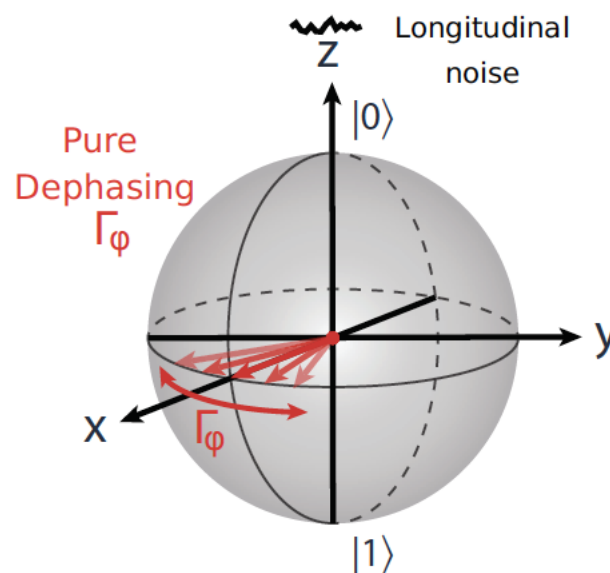
Relaxation (T1) and Decoherence (T2)

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \cos \frac{\theta}{2}|0\rangle + e^{i\phi} \sin \frac{\theta}{2}|1\rangle$$

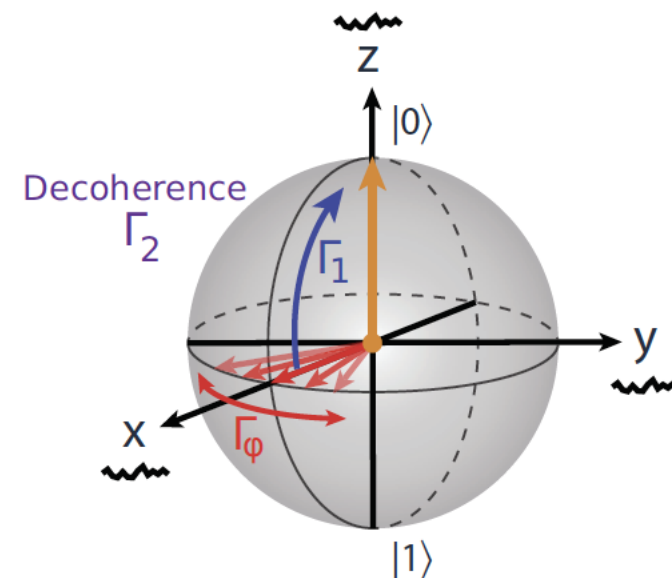
(b) Longitudinal relaxation



(c) Pure dephasing



(d) Transverse relaxation



Relaxation (T1) and Decoherence (T2)

The density matrix of the qubit evolves according to a **quantum master equation of Lindblad form**

$$\dot{\rho}_S = \underbrace{-i[H_S, \rho_S]}_{\text{Coherent dynamics}} + \underbrace{\sum_{j=1}^N \Gamma_j \left(L_j \rho_S L_j^\dagger - \frac{1}{2} \{L_j^\dagger L_j, \rho_S\} \right)}_{\text{Dissipative effects from the environment}}$$

Weak coupling
Markovian dynamics

Energy relaxation

$$L_1 = \sigma_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$
$$\gamma_1 = 1/T_1$$

Causes decay $|1\rangle \rightarrow |0\rangle$
and contributes to both
population and coherence decay.

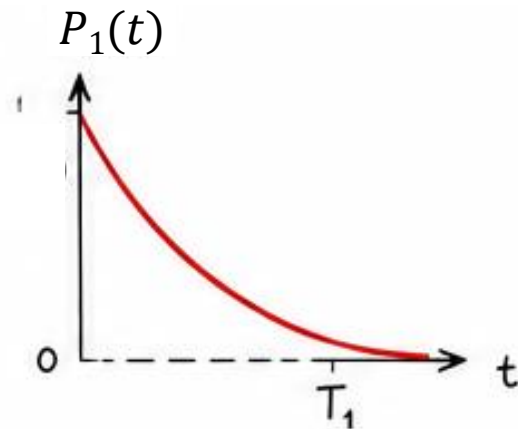
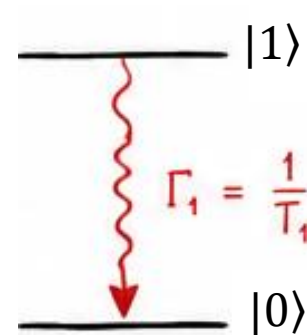
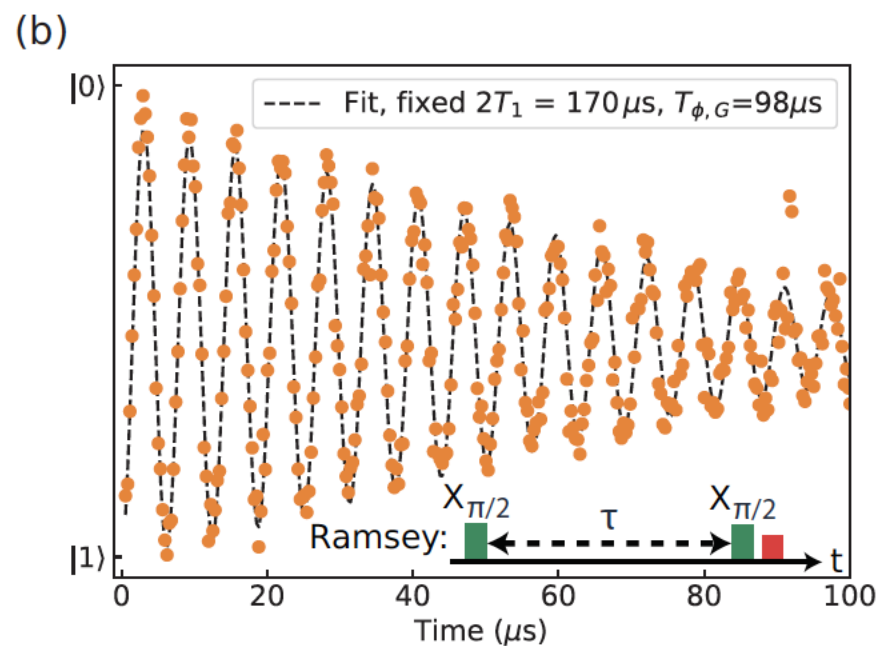
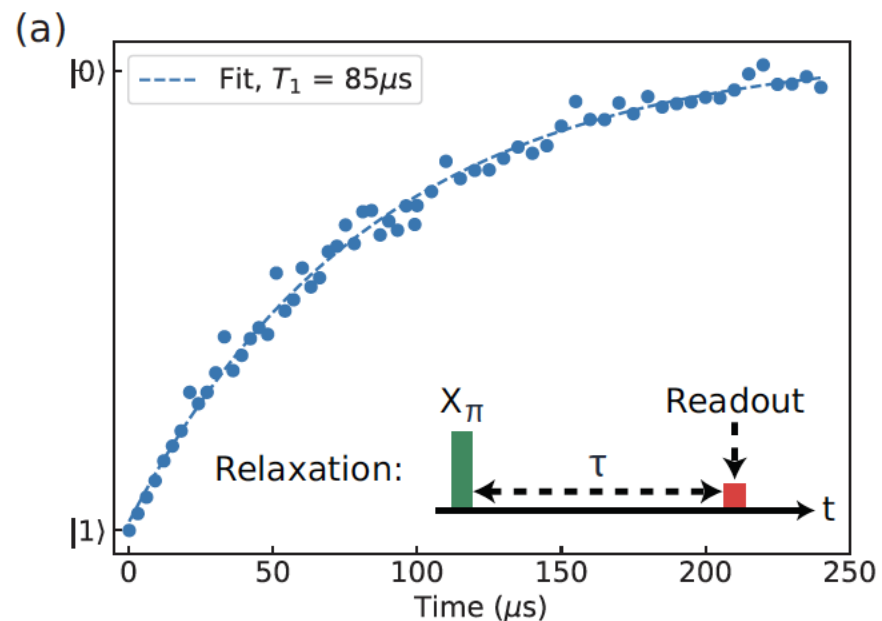
Pure dephasing

$$L_\phi = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$
$$\gamma_\phi = 1/T_\phi$$

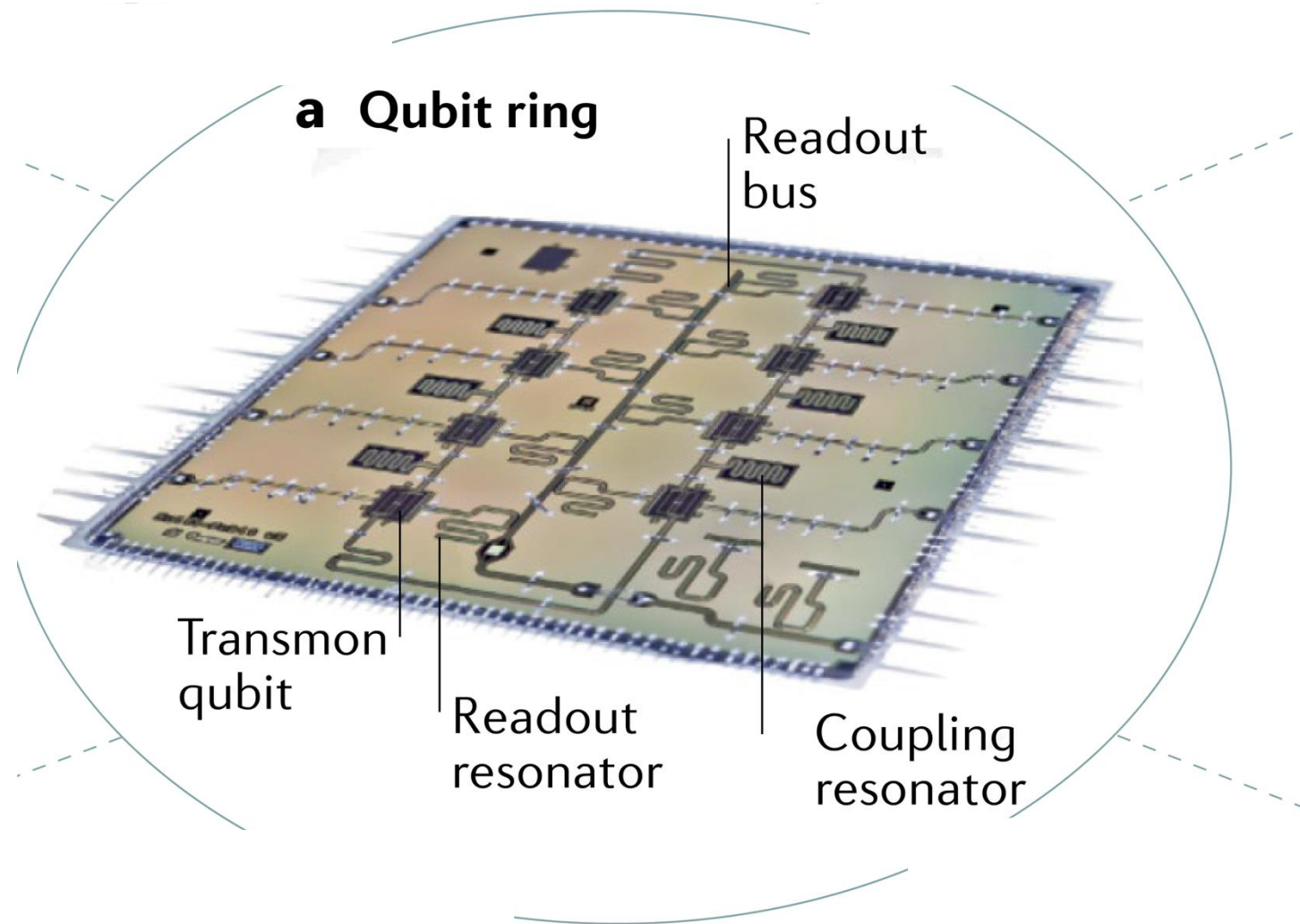
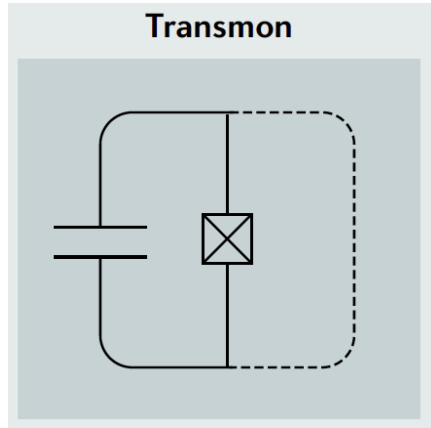
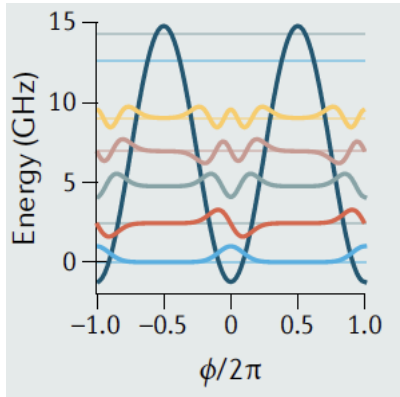
Causes random phase
fluctuations (no energy
exchange).

Relaxation (T1) and Decoherence (T2)

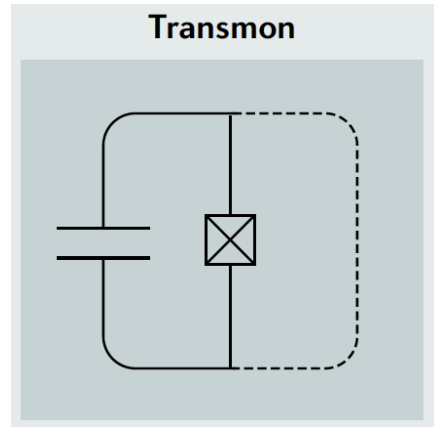
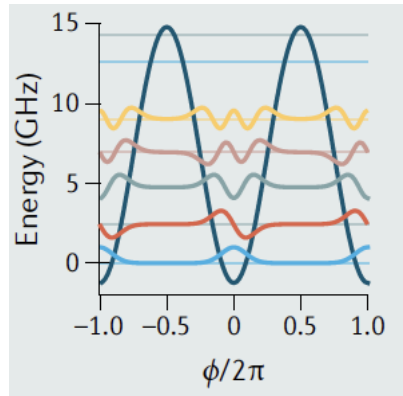
arXiv:1904.06560v5



A superconducting qubit (**the system**) interacts with an **environment** composed of different physical sources



A superconducting qubit (**the system**) interacts with an **environment** composed of different physical sources



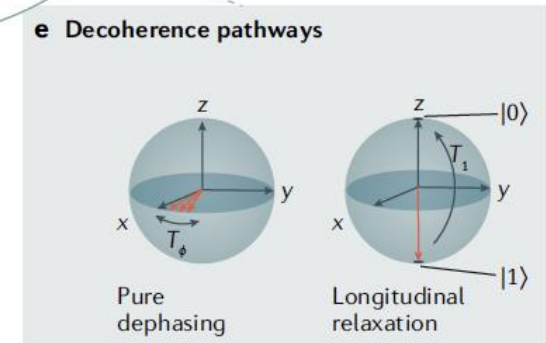
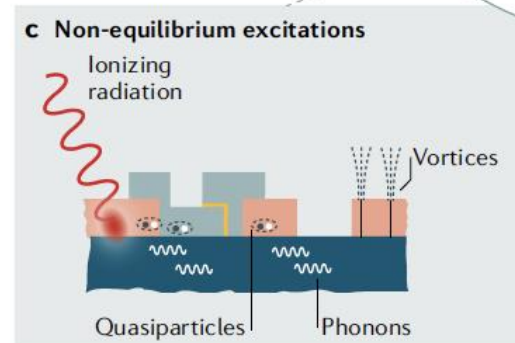
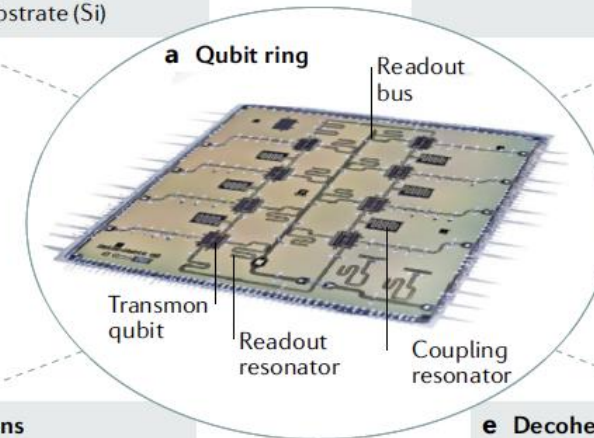
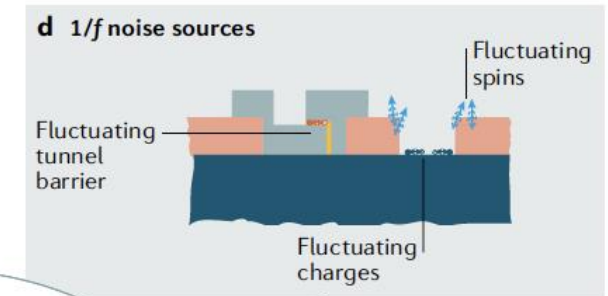
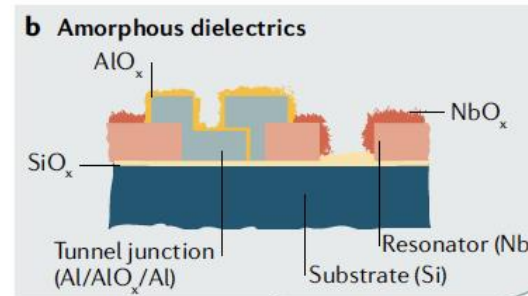
a | Optical image of a qubit ring consisting of eight transmons. Each qubit is coupled to its neighbour via a resonator.

b surface and interfacial defect

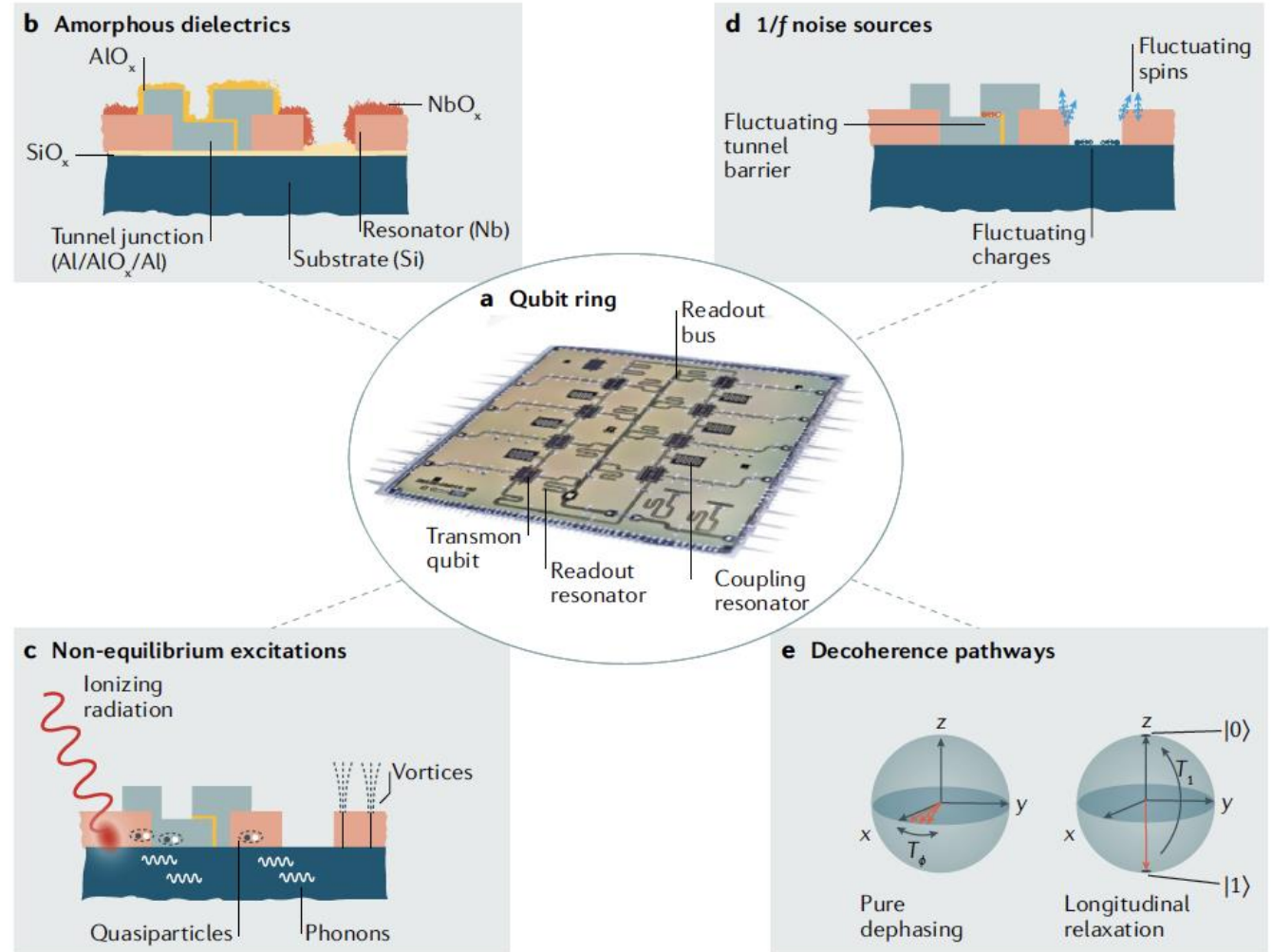
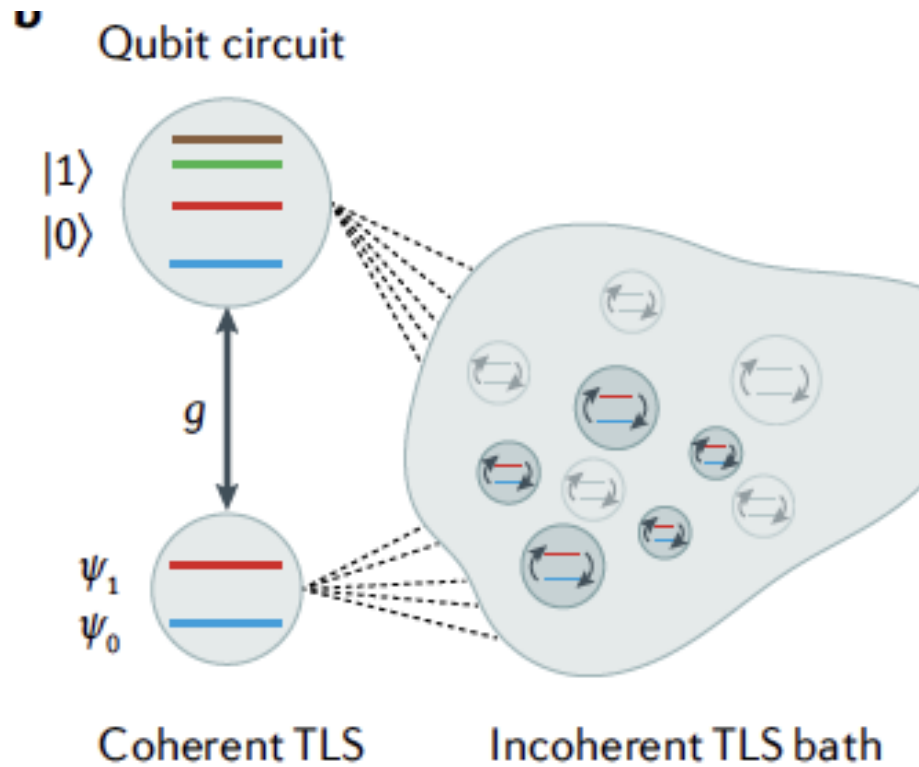
c | ionizing radiation and remnant magnetic fields

d | Microscopic charge and spin defects are believed to give rise to $1/f$ fluctuations

e | Fluctuations in the environment

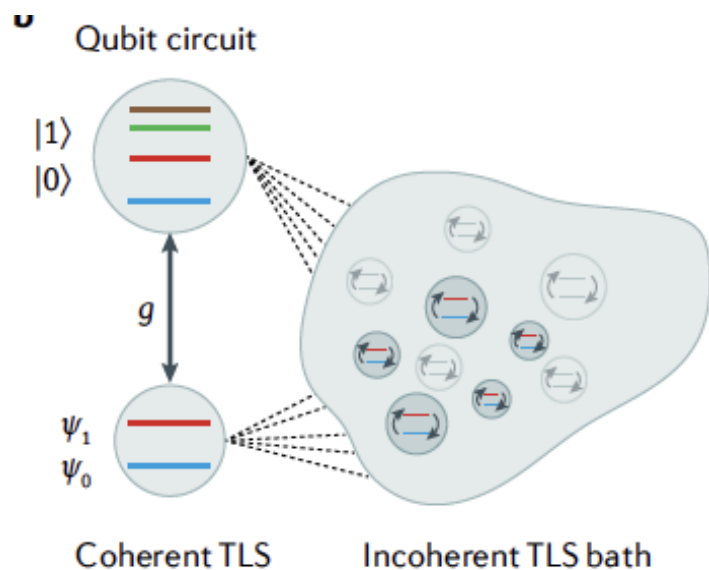


A superconducting qubit (**the system**) interacts with an **environment** composed of different physical sources

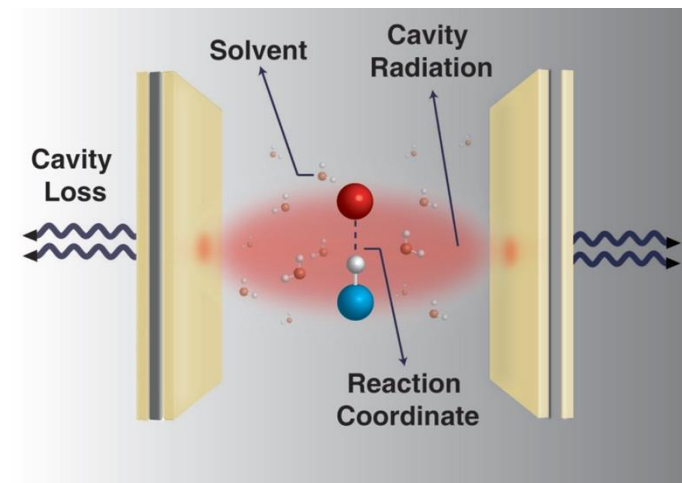
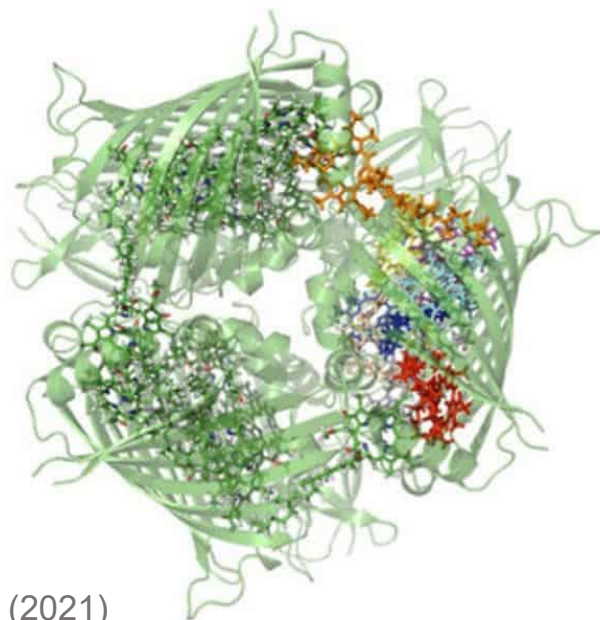


Ifran Siddiqi Nature Reviews
Materials 6, 875 (2021)

Quantum dissipative dynamics beyond weak system-bath coupling



Ifraan Siddiqi *Nature Reviews Materials* 6, 875 (2021)



W Ying, P Huo *Communications Materials* 5, 110 (2024)

Numerically exact tools (MC, path integrals, HEOM, wave function methods)

Perturbative tools (could be on a transformed Hamiltonian) – Markovian embedding

Mixed quantum-classical methods

Outline: Today's talk

○ Introduction

- About myself
- Intro: Open Quantum Systems
- T1 and T2 timescale and boring quantum dynamics

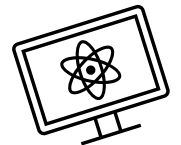


○ **Strong coupling:** Nonadditive effects



Slow down - separation of relaxation timescale

○ **Geometric heterogeneity:** disorder with dephasing leading to **long lived dynamics**



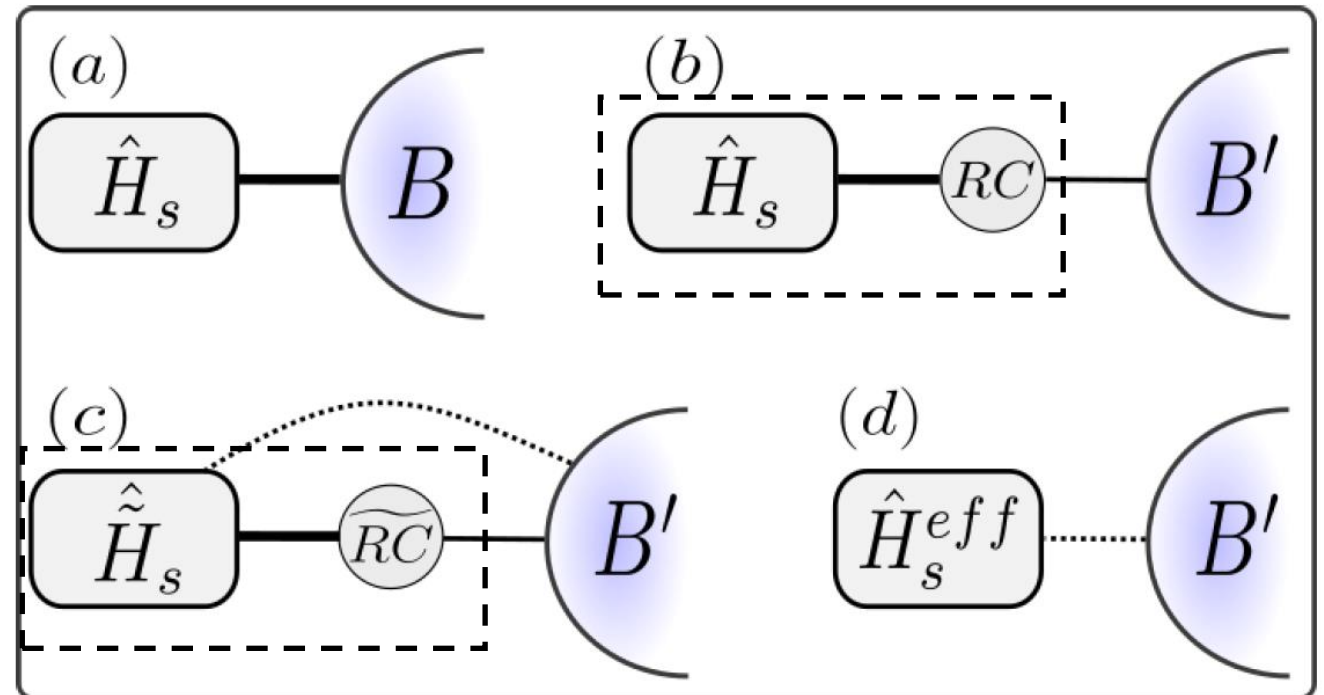
○ Summary

Open Access

Effective-Hamiltonian Theory of Open Quantum Systems at Strong Coupling

Nicholas Anto-Sztrikacs, Ahsan Nazir, and Dvira Segal
PRX Quantum **4**, 020307 – Published 13 April 2023

- (a) Begin with a generic open quantum system
- (b) Extend the system's Hilbert space (exact)
- (c) Rotation of frame (exact)
- (d) Truncation of the Hilbert space (approximation)

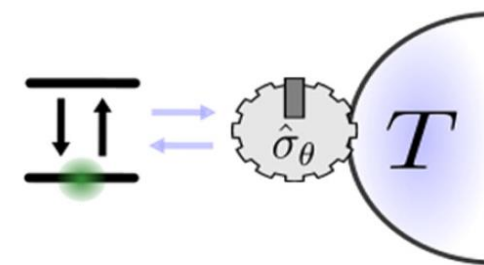


(1)

PRX QUANTUM 4, 020307 (2023)

Effective-Hamiltonian Theory of Open Quantum Systems at Strong Coupling

Nicholas Anto-Sztrikacs,¹ Ahsan Nazir², and Dvira Segal^{1,3,*}



(2)

PHYSICAL REVIEW B 110, 174304 (2024)

Open quantum systems with noncommuting coupling operators: An analytic approach

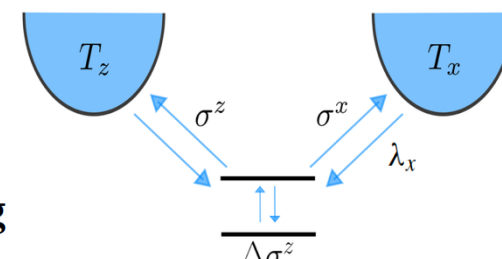
Jakub Garwoła¹ and Dvira Segal^{1,2,*}

PHYSICAL REVIEW A 112, 022209 (2025)

(3)

Suppression of decoherence dynamics by a dissipative bath at strong coupling

Jitian Chen^{1,*}, Jakub Garwoła^{2,†} and Dvira Segal^{1,2,†}

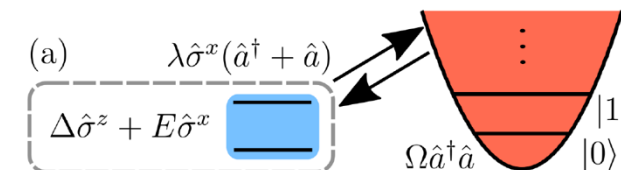


PHYSICAL REVIEW RESEARCH 8, L022039 (2026)

(4)

Compact cavity-dressed Hamiltonian framework at arbitrarily strong light-matter coupling

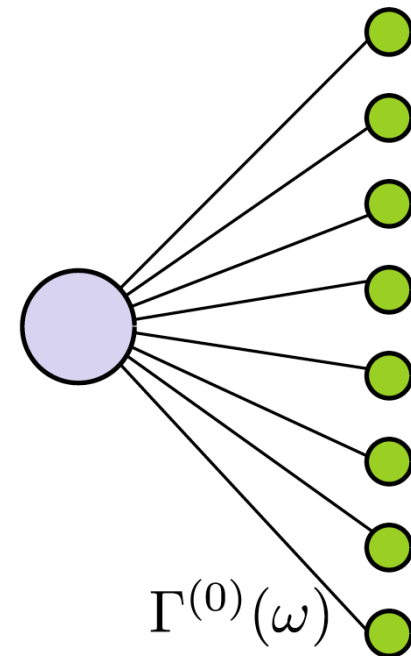
Jakub Garwoła^{1,*} and Dvira Segal^{1,2,†}



The Reaction Coordinate (RC) mapping

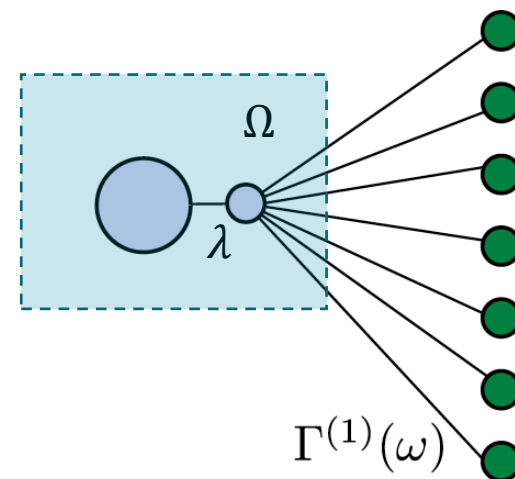
$$H = H_S + S \sum_k \left(h_k a_k + h_k^* a_k^\dagger \right) + \sum_k \omega_k a_k^\dagger a_k$$

$$\Gamma^{(0)}(\omega) = 2\pi \sum_k |h_k|^2 \delta(\omega - \omega_k)$$



$$H = H_S + \lambda S(b + b^\dagger) + \Omega b^\dagger b + (b + b^\dagger) \sum_k \tilde{H}_k \left(H_k b_k + H_k^* b_k^\dagger \right) + \sum_k \tilde{\Omega}_k b_k^\dagger b_k$$

$$\Gamma^{(1)}(\omega) = 2\pi \sum_k \tilde{|H_k|^2} \delta(\omega - \tilde{\Omega}_k)$$



**Strong S-B
couplings**

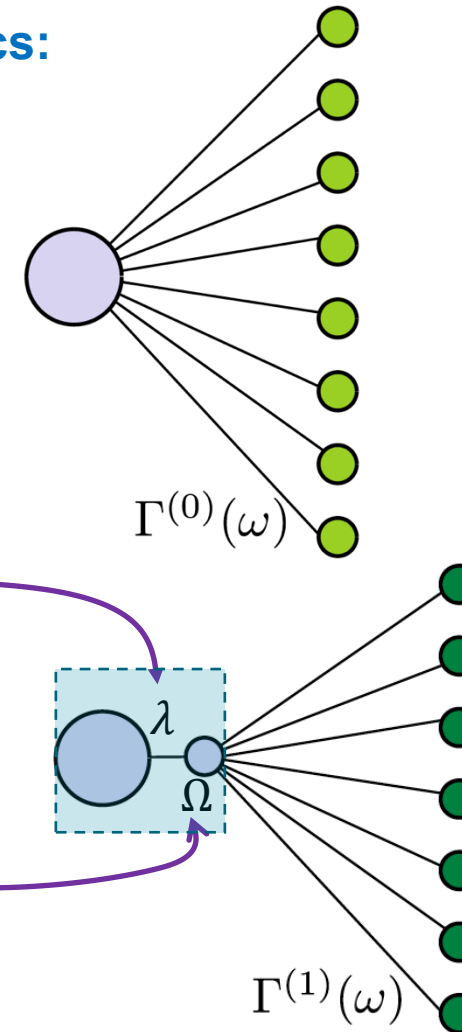
Strong coupling effects in Quantum thermodynamics: The Reaction Coordinate mapping

$$\Gamma^{(0)}(\omega) = 2\pi \sum_k |h_k|^2 \delta(\omega - \omega_k)$$

$$\lambda^2 = \frac{1}{2\pi\Omega} \int_0^\infty \omega \Gamma^{(0)}(\omega) d\omega$$

$$\Omega^2 = \frac{\int_0^\infty \omega^3 \Gamma^{(0)}(\omega) d\omega}{\int_0^\infty \omega \Gamma^{(0)}(\omega) d\omega}$$

$$\Gamma^{(1)}(\omega) = \frac{4\lambda^2 \Gamma^{(0)}(\omega)}{\left[\frac{1}{\pi} \mathcal{P} \int \frac{\Gamma^{(0)}(\omega')}{\omega' - \omega} d\omega' \right]^2 + [\Gamma^{(0)}(\omega)]^2}$$

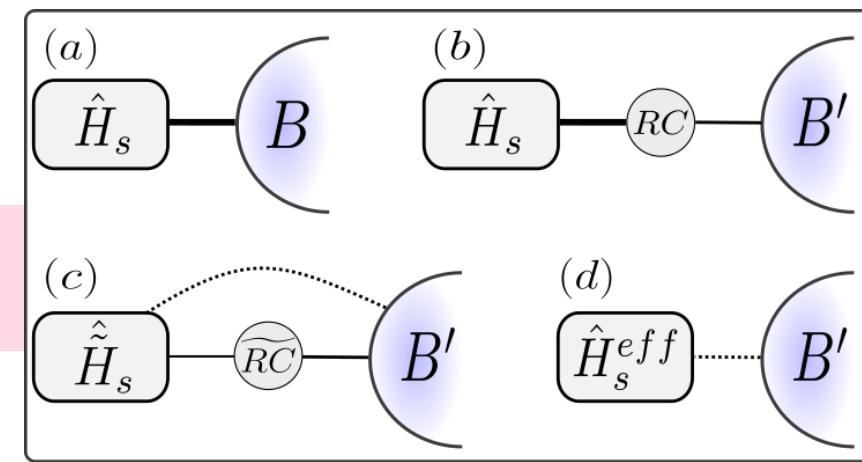


A. Nazir and G. Schaller,
edited by F. Binder, L.A.
Correa, C. Gogolin,
J. Anders, and G. Adesso
(Springer 2019), 551

(a) Model

$$\hat{H} = \hat{H}_S + \hat{H}_B + \hat{H}_I$$

$$= \hat{H}_S + \sum_k \nu_k \hat{c}_k^\dagger \hat{c}_k + \hat{S} \sum_k t_k (\hat{c}_k^\dagger + \hat{c}_k)$$



(b) Reaction coordinate mapping

$$\hat{H}_{\text{RC}} = \hat{H}_S + \Omega \hat{a}^\dagger \hat{a} + \lambda \hat{S} (\hat{a}^\dagger + \hat{a}) + \sum_k f_k (\hat{a}^\dagger + \hat{a}) (\hat{b}_k^\dagger + \hat{b}_k) + \sum_k \omega_k \hat{b}_k^\dagger \hat{b}_k$$

(c) Polaron shift

$$\hat{U}_P = e^{\frac{\lambda}{\Omega} (\hat{a}^\dagger - \hat{a}) \hat{S}}$$

$$\hat{\tilde{H}} = \hat{\tilde{H}}_S + \Omega \hat{a}^\dagger \hat{a} - \frac{\lambda^2}{\Omega} \hat{S}^2 + \sum_k f_k \left(\hat{a}^\dagger + \hat{a} - \frac{2\lambda}{\Omega} \hat{S} \right) (\hat{b}_k^\dagger + \hat{b}_k) + \sum_k \omega_k \hat{b}_k^\dagger \hat{b}_k$$

(d) Truncation via projection to the ground state of the RC, $\hat{Q}_0 = |0\rangle\langle 0|$:

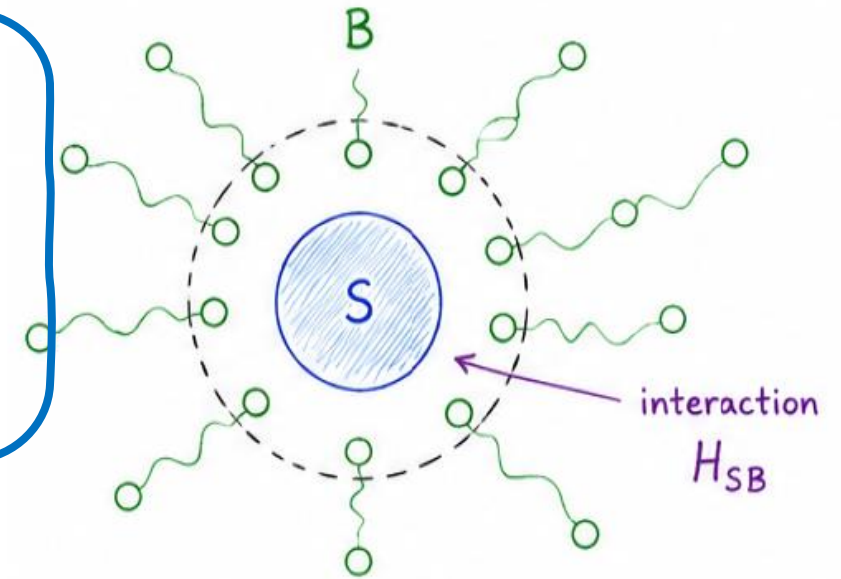
$$\hat{H}^{\text{eff}} = \hat{Q}_0 \hat{\tilde{H}}_S \hat{Q}_0 - \frac{\lambda^2}{\Omega} \hat{S}^2 - \hat{S} \sum_k \frac{2\lambda f_k}{\Omega} (\hat{b}_k^\dagger + \hat{b}_k) + \sum_k \omega_k \hat{b}_k^\dagger \hat{b}_k$$

Example: A three level model

$$\hat{H} = \hat{H}_S + \hat{H}_I + \hat{H}_B$$

$$\hat{H}_S = (v - \Delta) |2\rangle \langle 2| + v |3\rangle \langle 3|$$

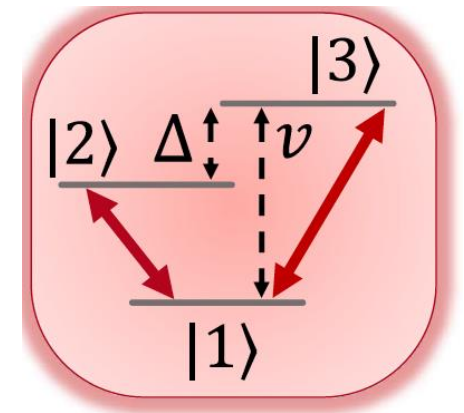
$$\hat{H}_I = \hat{S} \sum_k t_k (\hat{c}_k^\dagger + \hat{c}_k), \quad \hat{H}_B = \sum_k \nu_k \hat{c}_k^\dagger \hat{c}_k$$



We are interested in the dynamics of S, not the full S+B.

$$\hat{H}^{\text{eff}} = \hat{H}_S^{\text{eff}}(\lambda, \Omega) - \hat{S} \sum_k \frac{2\lambda f_k}{\Omega} (\hat{b}_k^\dagger + \hat{b}_k) + \sum_k \omega_k \hat{b}_k^\dagger \hat{b}_k$$

$$\hat{H}_S^{\text{eff}}(\lambda, \Omega) = \langle 0 | \hat{U}_P \hat{H}_S \hat{U}_P^\dagger | 0 \rangle - \frac{\lambda^2}{\Omega} \hat{S}^2$$



$$\dot{\rho}_S = -i [H_S, \rho_S] + \sum_{j=1}^N \Gamma_j \left(L_j \rho_S L_j^\dagger - \frac{1}{2} \left\{ L_j^\dagger L_j, \rho_S \right\} \right)$$

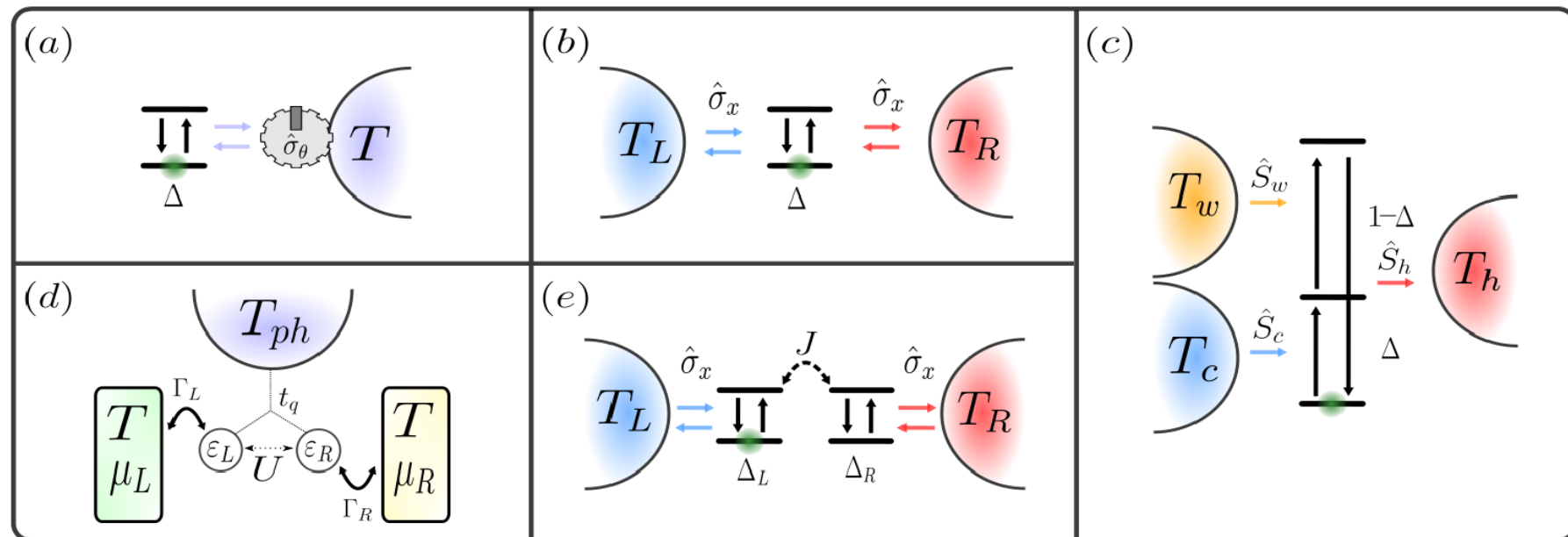
Coherent
dynamics

Dissipative effects from the
environment

Open Access

Effective-Hamiltonian Theory of Open Quantum Systems at Strong Coupling

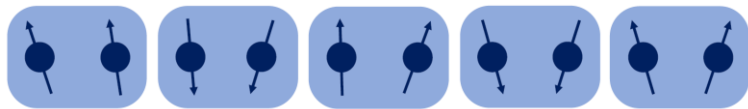
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PRX Quantum **4**, 020307 – Published 13 April 2023

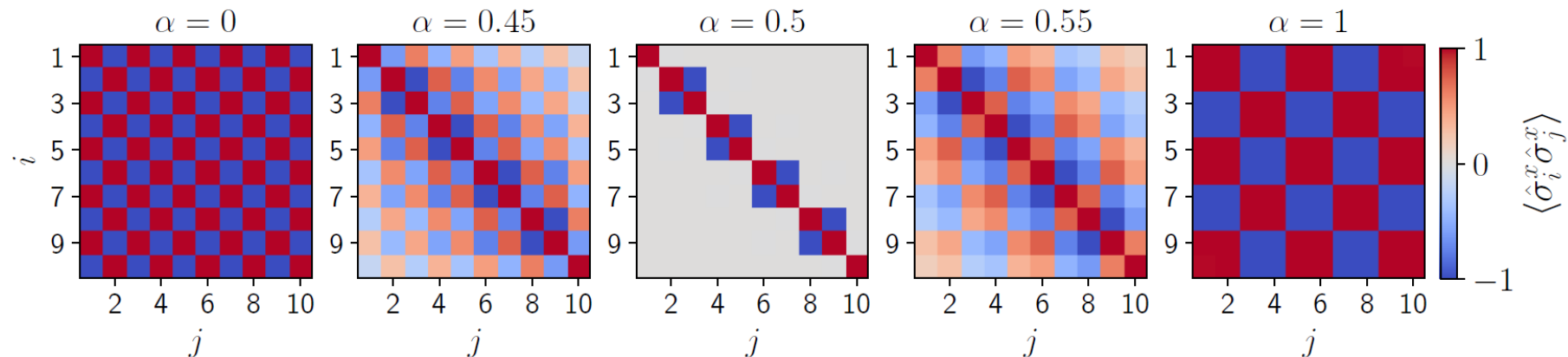
Bath-Engineering Magnetic Order in Quantum Spin Chains: An Analytic Mapping Approach

Brett Min,^{1,*} Nicholas Anto-Sztrikacs,¹ Marlon Brenes¹ and Dvira Segal^{2,1,†}

AFM Ising model with *pairwise* coupling

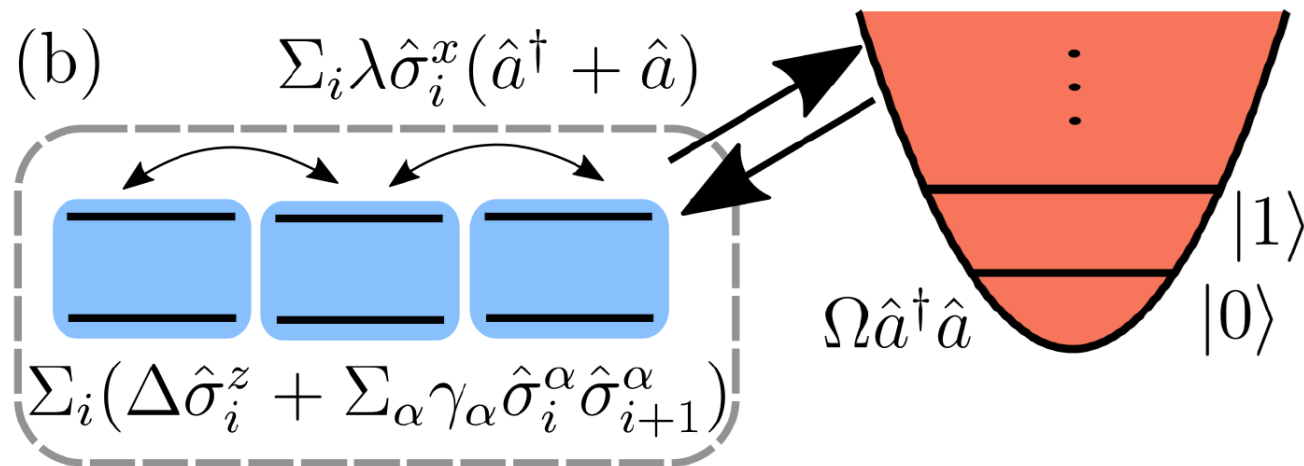
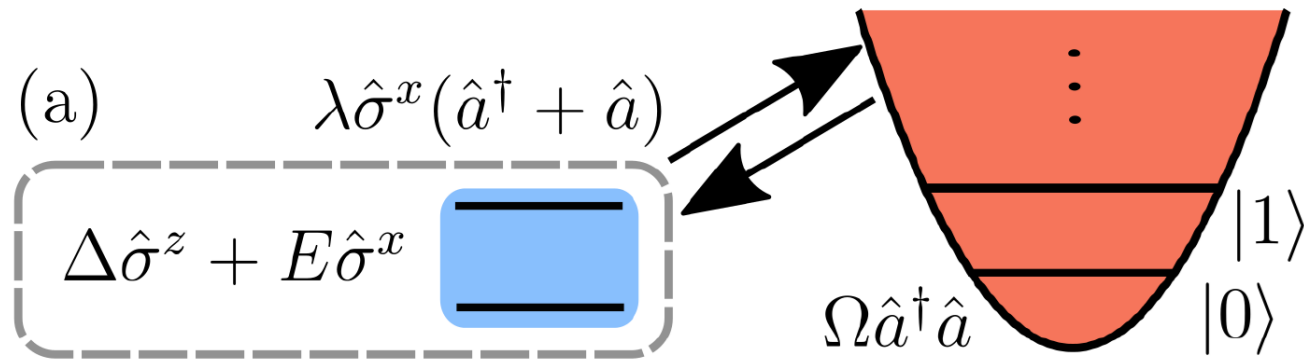


AFM to extended Neel



Compact cavity-dressed Hamiltonian framework at arbitrarily strong light-matter coupling

Jakub Garwoła, Dvira Segal



Compact cavity-dressed Hamiltonian framework at arbitrarily strong light-matter coupling

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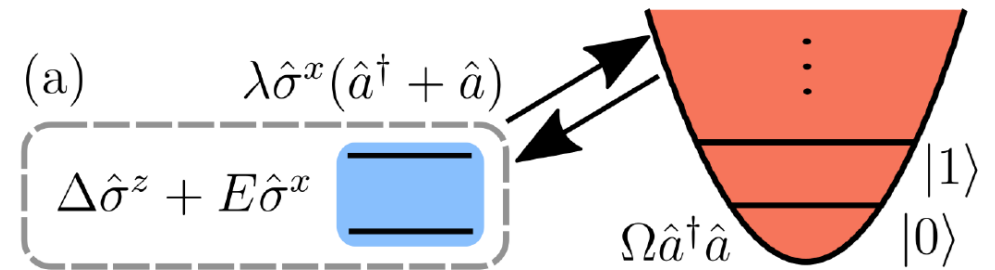
$$\hat{H} = \Delta \hat{\sigma}^z + \Omega \hat{a}^\dagger \hat{a} + \lambda \hat{\sigma}^x (\hat{a}^\dagger + \hat{a})$$

Cavity mode Light-matter interaction

$$\hat{H}_{(2)}^{\text{cdh}} = \begin{pmatrix} \tilde{\Delta} \hat{\sigma}^z - \Omega \epsilon^2 \hat{1} & 2\tilde{\Delta} \epsilon i \hat{\sigma}^y \\ -2\tilde{\Delta} \epsilon i \hat{\sigma}^y & \tilde{\Delta} (1 - 4\epsilon^2) \hat{\sigma}^z + \Omega (1 - \epsilon^2) \hat{1} \end{pmatrix}$$

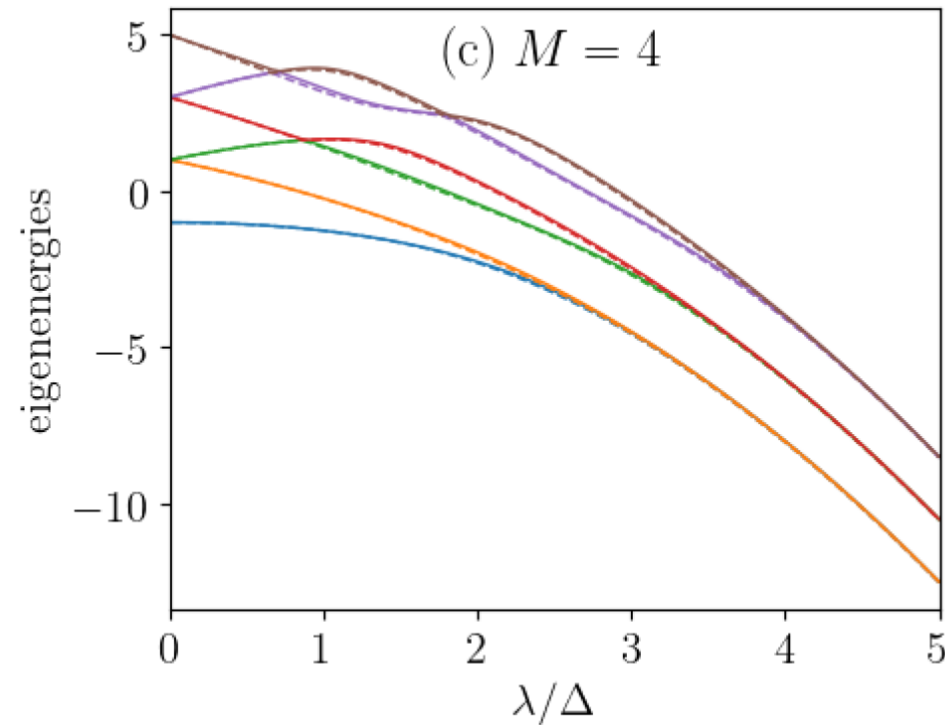
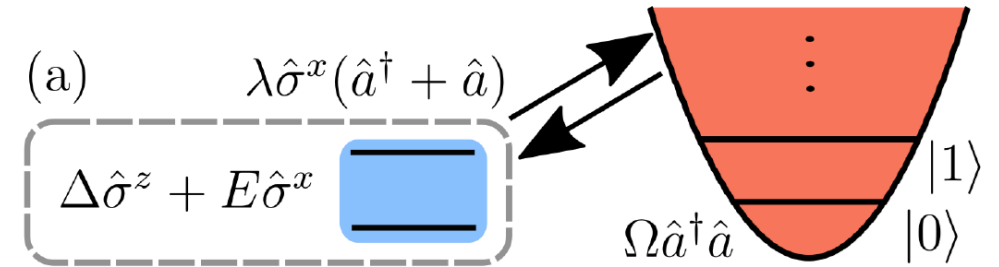
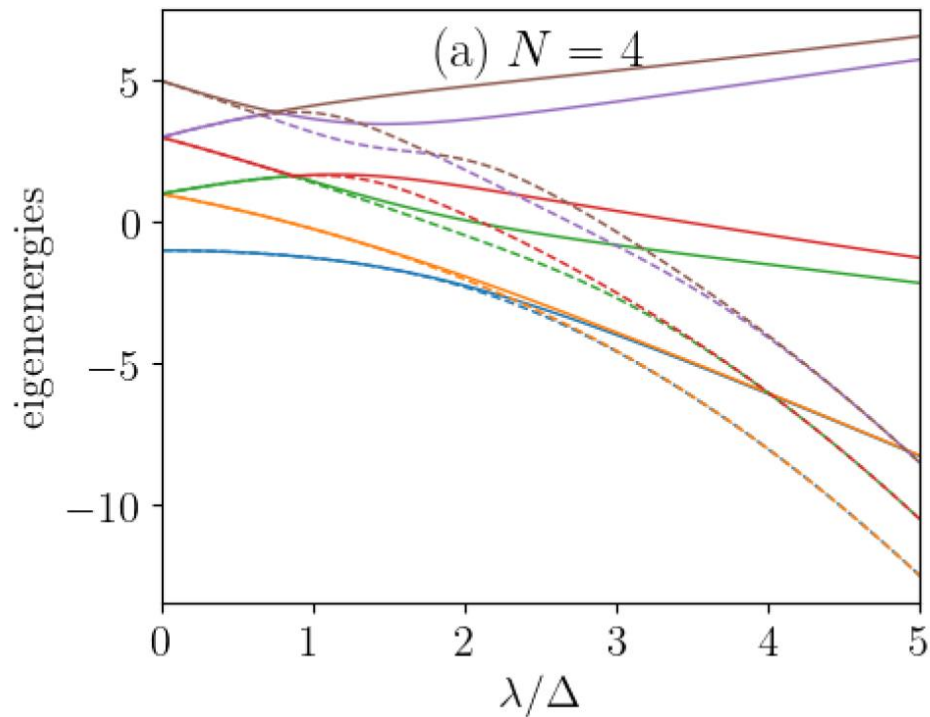
$$\epsilon = \lambda / \Omega$$

$$\tilde{\Delta} = \Delta e^{-2\epsilon^2}$$



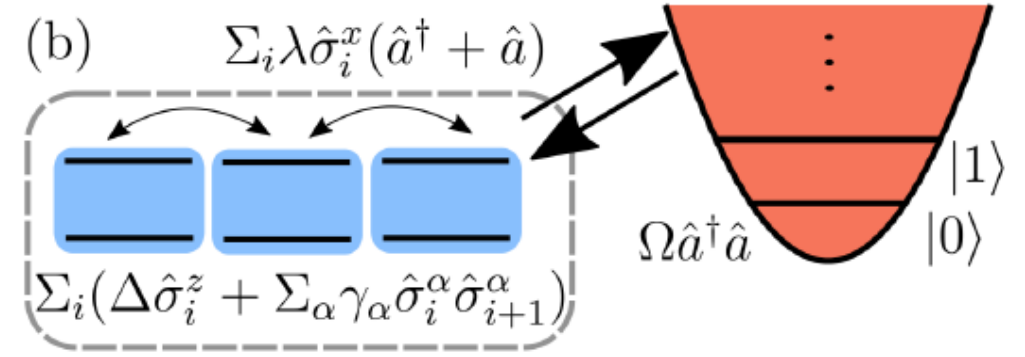
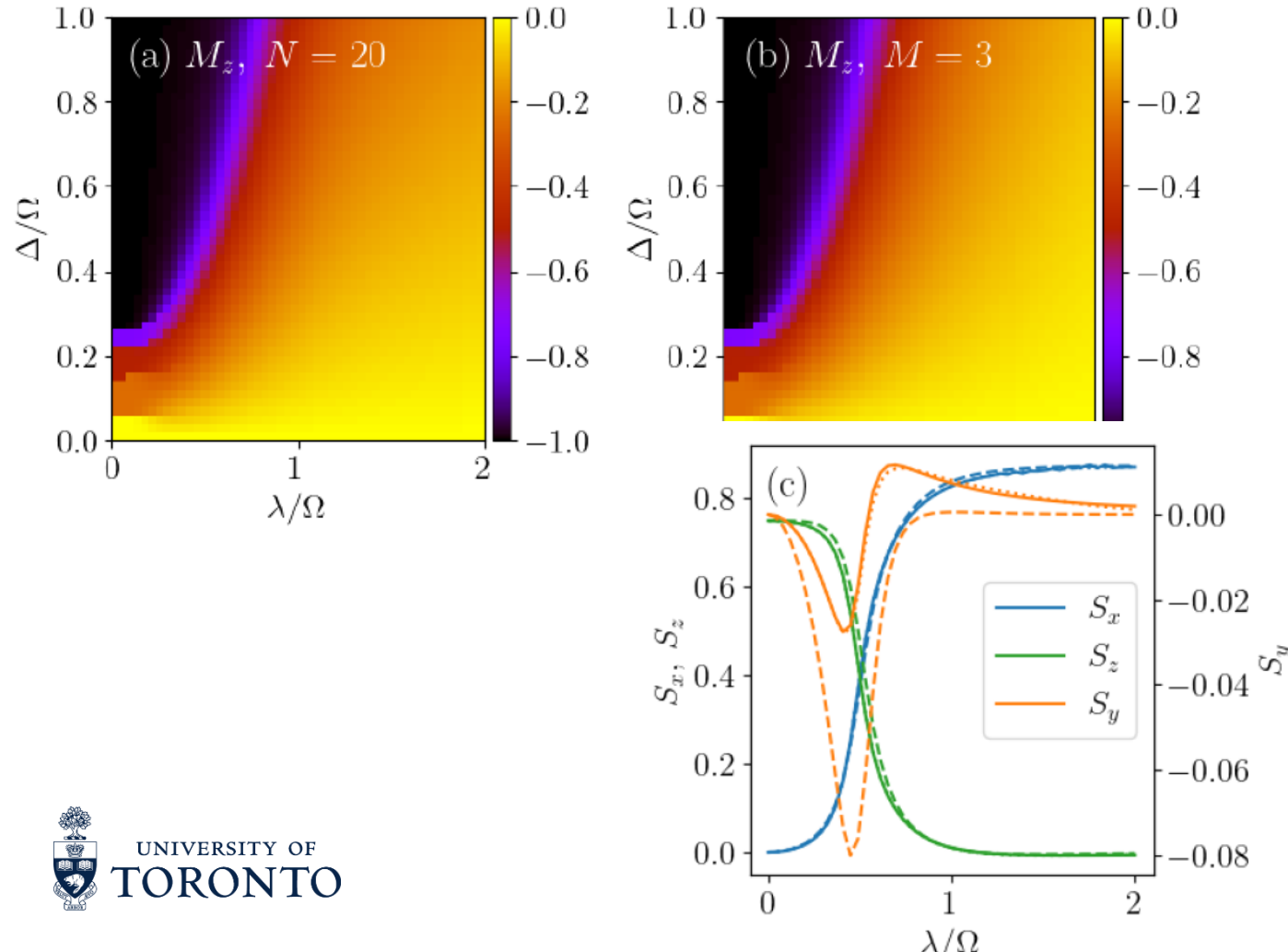
Compact cavity-dressed Hamiltonian framework at arbitrarily strong light-matter coupling

Jakub Garwoła, Dvira Segal



Compact cavity-dressed Hamiltonian framework at arbitrarily strong light-matter coupling

Jakub Garwoła, Dvira Segal



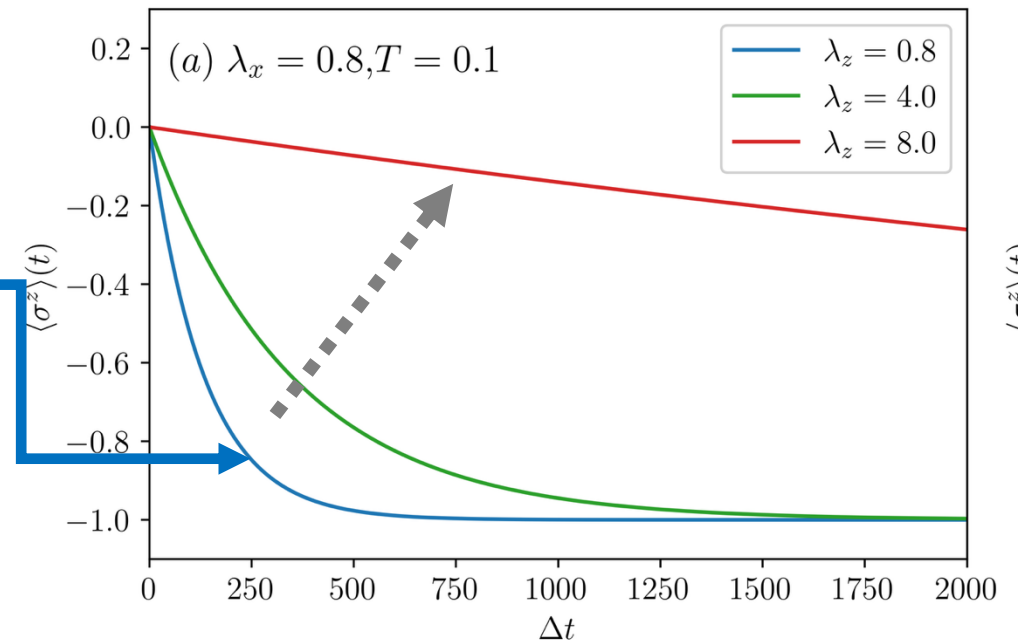
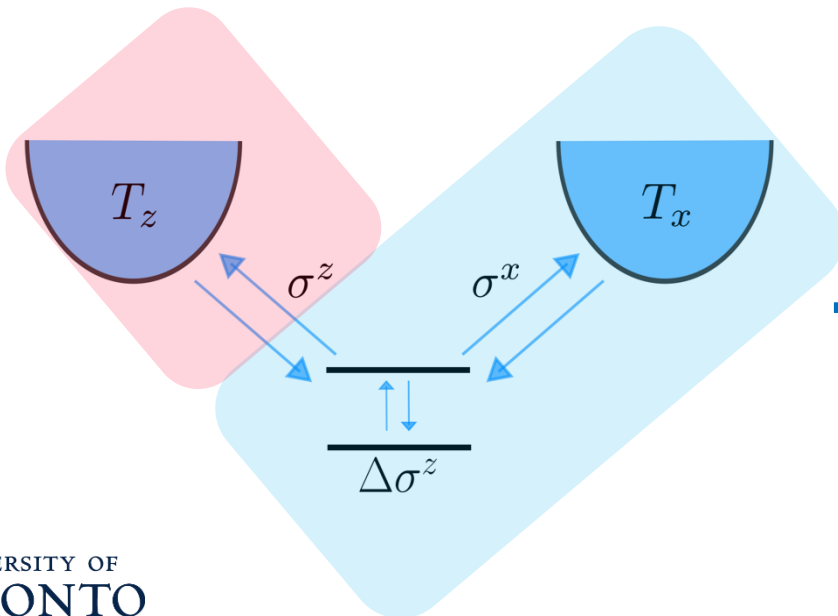
$$\hat{H} = \Delta \sum_{i=1}^L \hat{\sigma}_i^z + \Omega \hat{a}^{\dagger} \hat{a} + \sum_{i=1}^L \left(\frac{\lambda}{\sqrt{L}} \hat{\sigma}_i^x (\hat{a}^{\dagger} + \hat{a}) - \sum_{\alpha=x,y,z} \gamma_{\alpha} \hat{\sigma}_i^{\alpha} \hat{\sigma}_{i+1}^{\alpha} \right)$$

$$S_{\alpha} = \sum_{i,j} \langle \hat{\sigma}_i^{\alpha} \hat{\sigma}_j^{\alpha} \rangle_{GS} / L^2$$

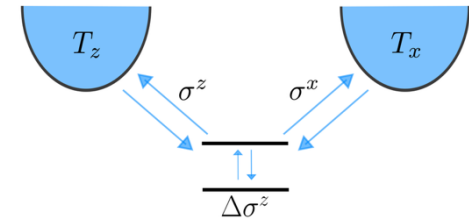
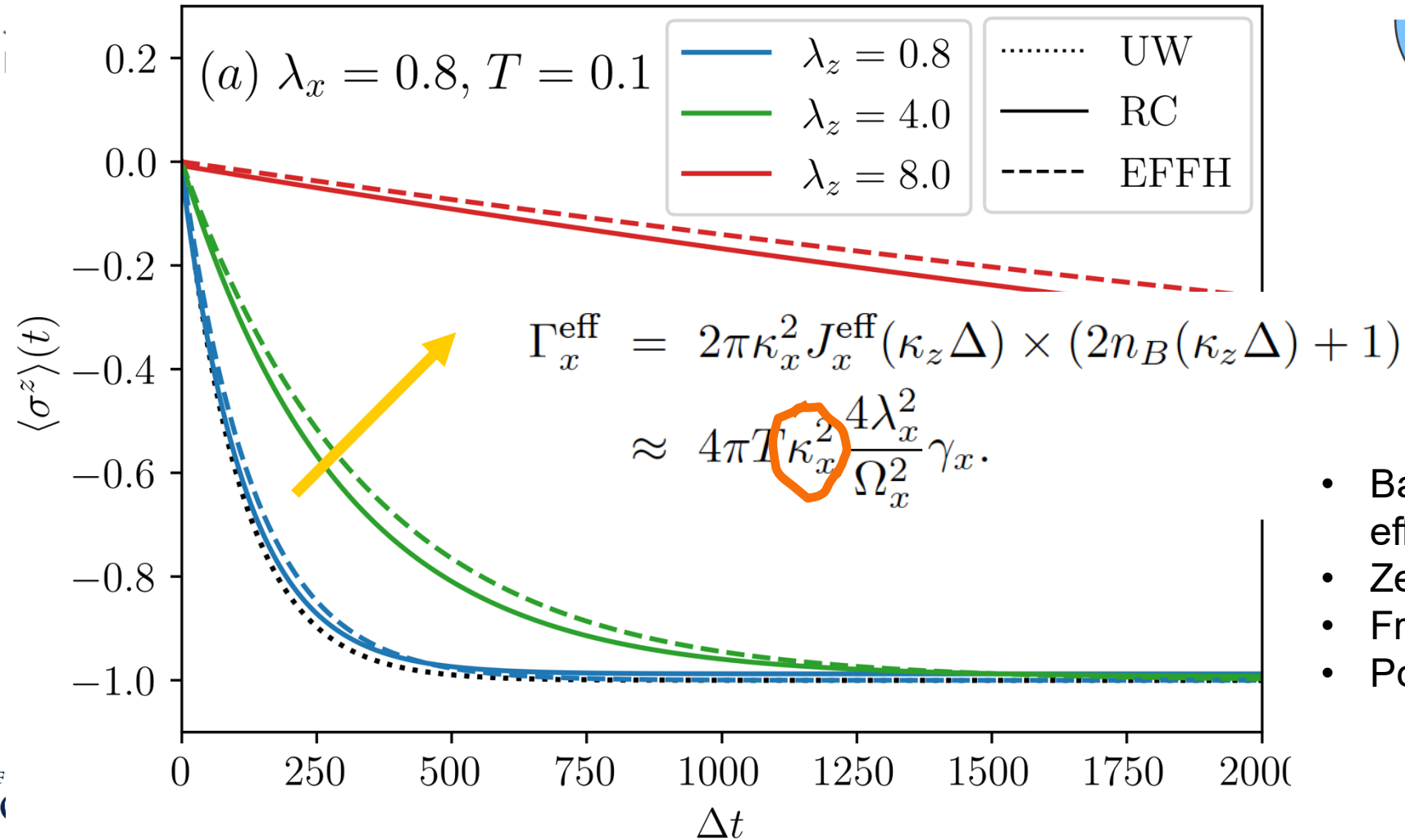
Open quantum systems with **noncommuting** coupling operators: An analytic approach

Jakub Garwoła¹ and Dvira Segal^{1,2,*}

We present an **analytic** approach to treat **open quantum systems strongly coupled to multiple** environments via **noncommuting** system operators: a prime example is a qubit concurrently coupled to both **decohering** and **dissipative** baths. For a spin impurity coupled to both dissipative and decohering environments, the effective Hamiltonian predicts the suppression of relaxation by decoherence, a phenomenon previously observed in simulations but lacking so far a theoretical foundation.



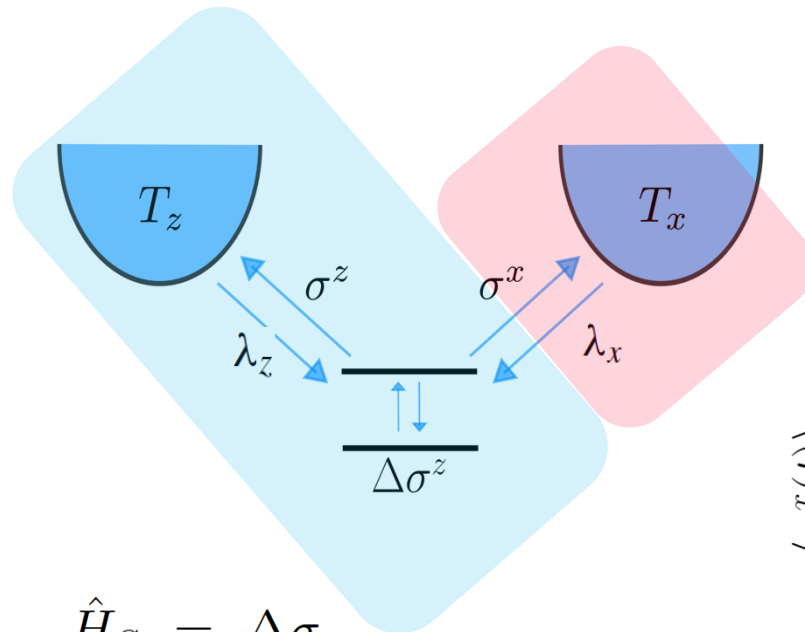
Open quantum systems with noncommuting coupling operators: An analytic approach



- Baths-cooperative effects
- Zeno effect
- Frustration
- Polaron undressing

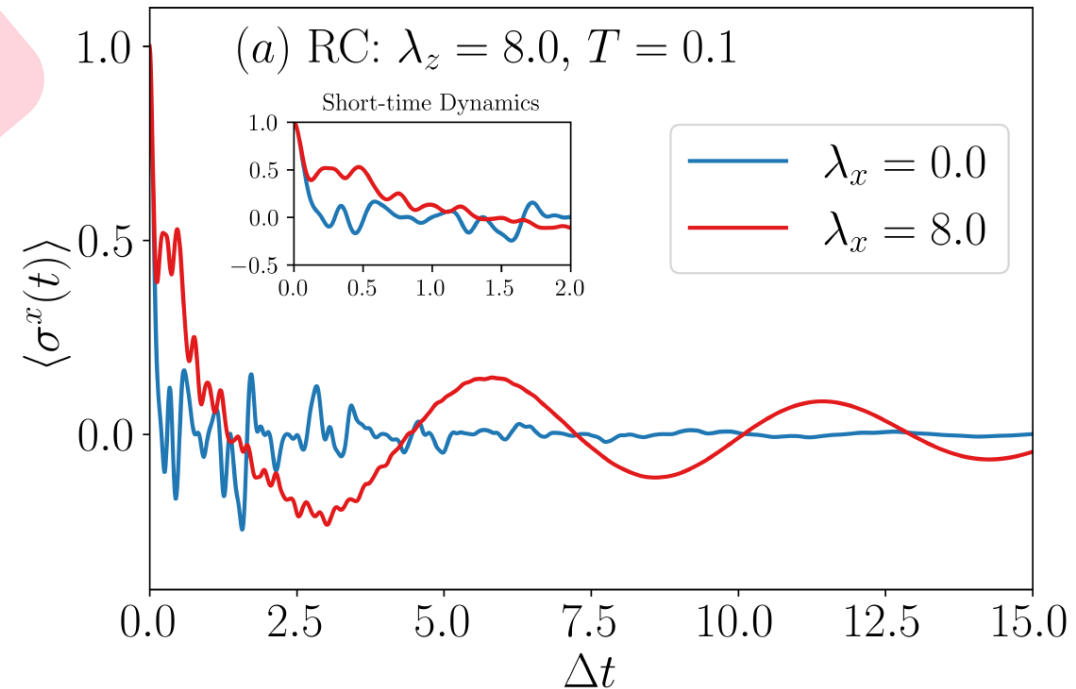
Suppression of decoherence dynamics by a dissipative bath at strong coupling

Jitian Chen^{1,*}, Jakub Garwoła^{2,*} and Dvira Segal^{1,2,§}



$$\begin{aligned}\hat{H}_S &= \Delta\sigma_z, \\ \hat{S}_z &= \hat{\sigma}_z, \quad \hat{S}_x = \hat{\sigma}_x.\end{aligned}$$

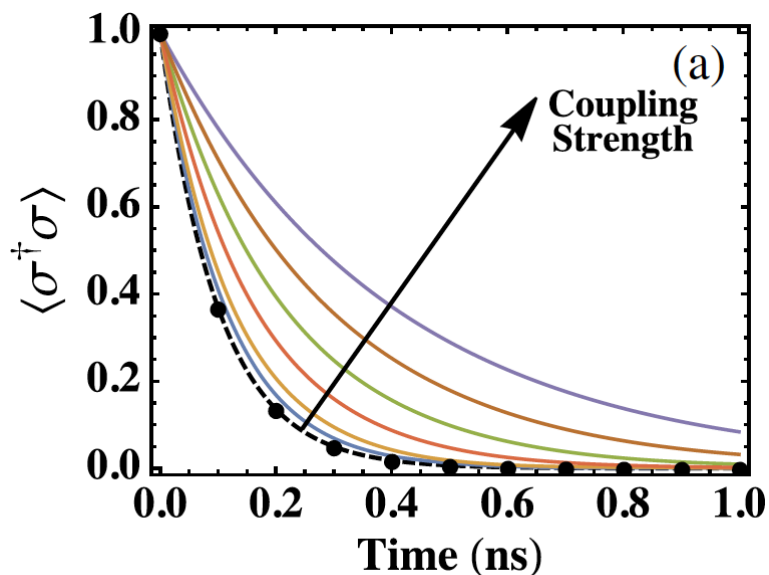
“Vice Versa”: Suppression of decoherence by relaxation



Environmental Nonadditivity and Franck-Condon physics in Nonequilibrium Quantum Systems

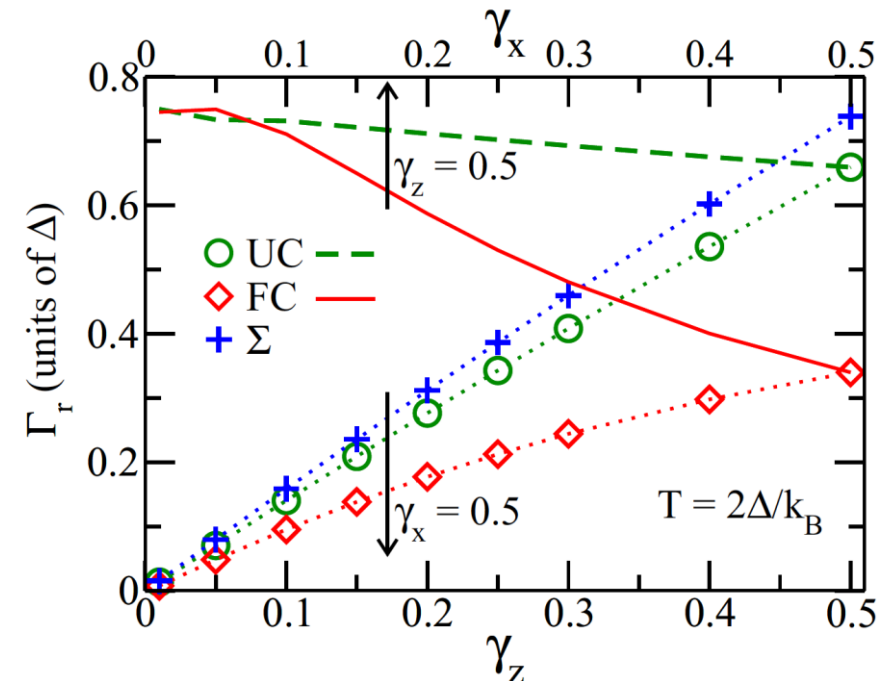
Henry Maguire, Jake Iles-Smith, Ahsan Nazir
Phys. Rev. Lett. **123**, 093601 (2019)

Reaction Coordinate simulations



Emitter population dynamics from the additive (dots) and nonadditive (solid) theories for increasing vibrational coupling strength

Influence functional Path integral



Suppressing relaxation through dephasing

T. Palm and P. Nalbach, PRA 103,022206 (2021)

We find that increasing purely dephasing fluctuations suppress increasing relaxation in the quantum system. .. These effects arise for medium to strong primary fluctuations already when the secondary fluctuations are weak due to their noncommuting coupling to the quantum system. Dephasing, in contrast, is increased by increasing any of the two fluctuations.

Readout-Induced Suppression and Enhancement of Superconducting Qubit Lifetimes

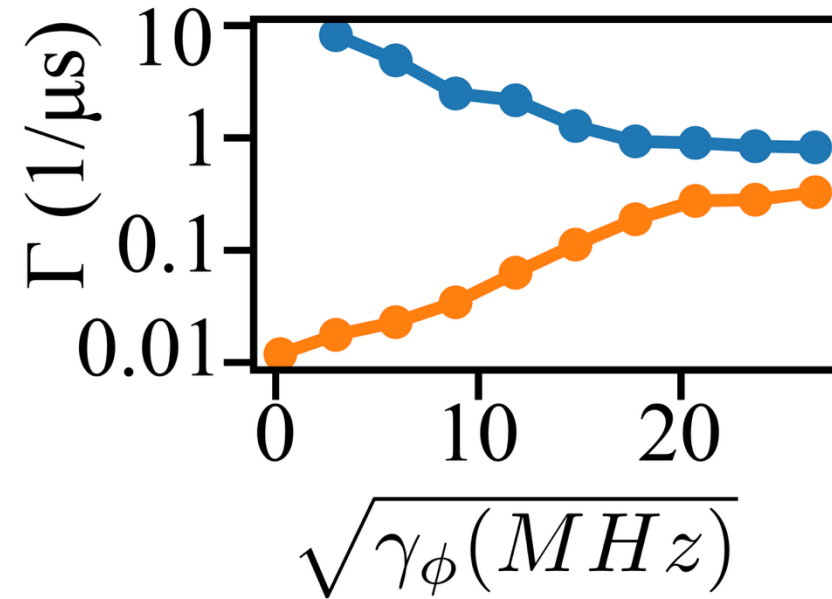
Ted Thorbeck^{1,*}, Zhihao Xiao², Archana Kamal², and Luke C. G. Govia³

¹*IBM Quantum, IBM T. J. Watson Research Center, Yorktown Heights, New York 10598, USA*

²*Department of Physics and Applied Physics, University of Massachusetts, Lowell, Massachusetts 01854, USA*

³*IBM Quantum, IBM Almaden Research Center, San Jose, California 95120, USA*

We accurately predict the change in lifetime during readout with a new self-consistent master equation that incorporates the modification to qubit relaxation due to measurement-induced dephasing. Moreover, we controllably demonstrate both the Zeno and anti-Zeno effects, which can explain both **suppression and the rarer enhancement of qubit lifetimes during readout.**



$$\Gamma(\bar{n}_r) = \int_{-\infty}^{\infty} \gamma_q(\omega) \frac{1}{\pi} \frac{\gamma_\phi(\bar{n}_r)}{\gamma_\phi(\bar{n}_r)^2 + (\omega - \tilde{\omega}_q(\bar{n}_r))^2} d\omega$$

Outline: Today's talk

○ Introduction

- About myself
- Intro: Open Quantum Systems
- T1 and T2 timescale and boring quantum dynamics

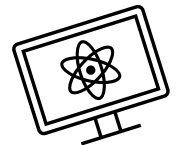


○ Strong coupling: Nonadditive effects



Slow down - separation of relaxation timescale

○ Geometric heterogeneity: disorder with dephasing leading to long lived dynamics



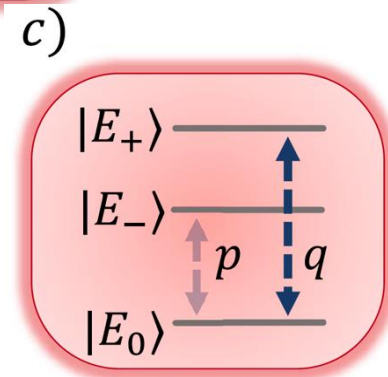
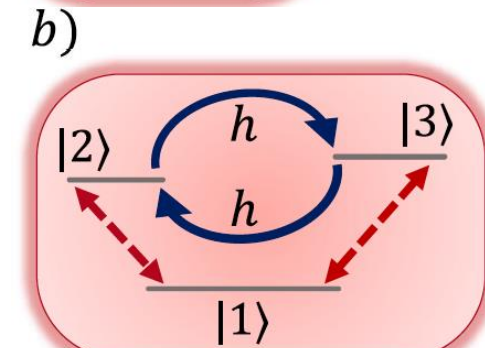
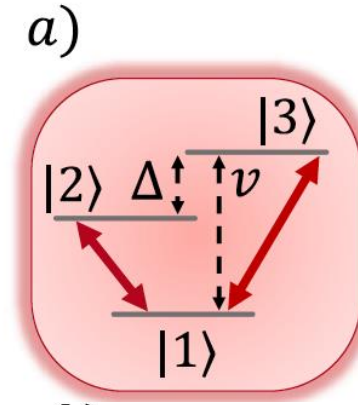
○ Summary

Separation of relaxation timescales via strong system–bath coupling: Dissipative three-level system as a case study

Brett Min, Matthew Gerry, Dvira Segal

$$\begin{aligned}\hat{H}_{\text{RC}} &= \hat{H}_S + \lambda \hat{S} (\hat{a}^\dagger + \hat{a}) + \Omega \hat{a}^\dagger \hat{a} \\ &+ (\hat{a}^\dagger + \hat{a}) \sum_k f_k (\hat{b}_k^\dagger + \hat{b}_k) + \sum_k \omega_k \hat{b}_k^\dagger \hat{b}_k \\ \hat{S} &= \frac{|1\rangle\langle 2| + |1\rangle\langle 3|}{\sqrt{2}} + \text{H.c.}\end{aligned}$$

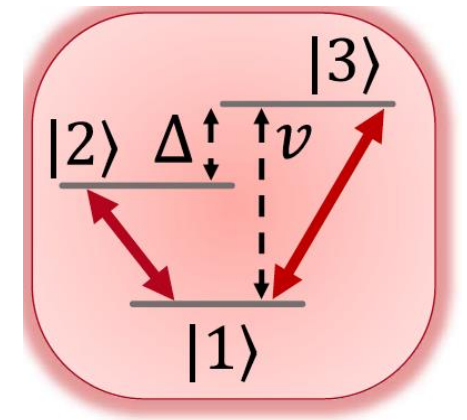
$$\begin{aligned}\hat{H}_E^{\text{eff}} &= \hat{H}_{S,E}^{\text{eff}} - \begin{pmatrix} 0 & p & q \\ p & 0 & 0 \\ q & 0 & 0 \end{pmatrix} \sum_k \frac{2\lambda f_k}{\Omega} (\hat{b}_k^\dagger + \hat{b}_k) \\ &+ \sum_k \omega_k \hat{b}_k^\dagger \hat{b}_k, \\ \hat{H}_S^{\text{eff}} &= \begin{pmatrix} E_0 & 0 & 0 \\ 0 & l + w & h \\ 0 & h & l - w \end{pmatrix} \quad \begin{aligned} p^2 &\approx \frac{\Omega^2 \Delta^2 \exp(-\frac{\lambda^2}{\Omega^2})}{4\lambda^4}, \\ q^2 &\approx 1 - \frac{\Omega^2 \Delta^2 \exp(-\frac{\lambda^2}{\Omega^2})}{4\lambda^4} \end{aligned}\end{aligned}$$



Separation of relaxation timescales via strong system–bath coupling: Dissipative three-level system as a case study

Brett Min, Matthew Gerry, Dvira Segal

$$\dot{\rho}_S = -i [H_S, \rho_S] + \sum_{j=1}^N \Gamma_j \left(L_j \rho_S L_j^\dagger - \frac{1}{2} \{ L_j^\dagger L_j, \rho_S \} \right)$$



Eigenvalues of the Liouvillian are the relaxation rates

Intuitive

$$\tau_1 \propto \left(\frac{\Omega}{\lambda} \right)^2 \frac{1}{T}$$

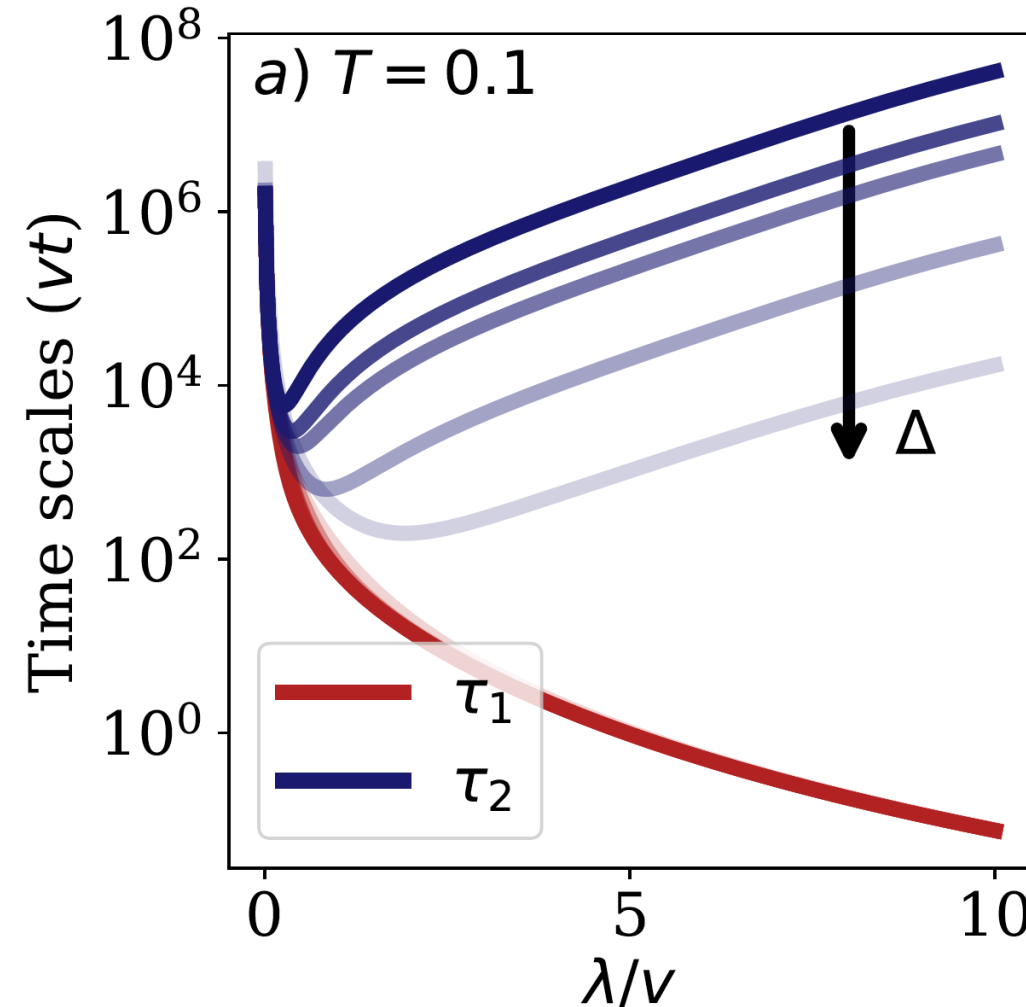
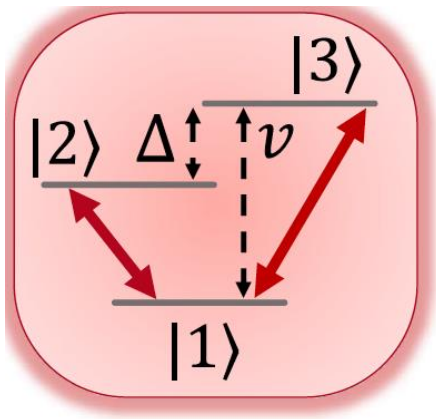
$$\tau_2 \propto \left(\frac{\lambda}{\Delta} \right)^2 \frac{1}{T} e^{\lambda^2/\Omega^2}$$

Counter intuitive!

$$\dot{\rho}_S = -i[H_S, \rho_S] + \sum_{j=1}^N \Gamma_j \left(L_j \rho_S L_j^\dagger - \frac{1}{2} \{ L_j^\dagger L_j, \rho_S \} \right)$$

Separation of relaxation timescales via strong system–bath coupling: Dissipative three-level system as a case study

Brett Min, Matthew Gerry, Dvira Segal



$$\tau_2 \propto \left(\frac{\lambda}{\Delta} \right)^2 \frac{1}{T} e^{\lambda^2/\Omega^2}$$

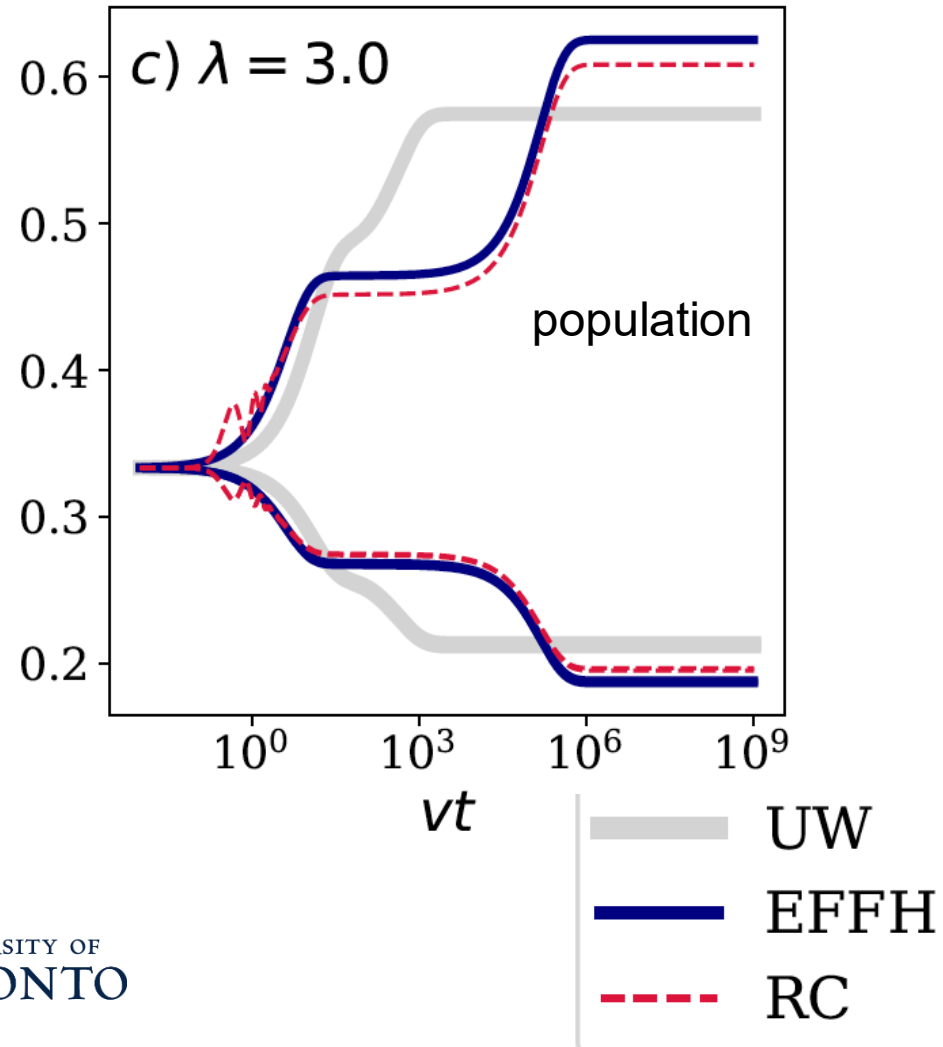
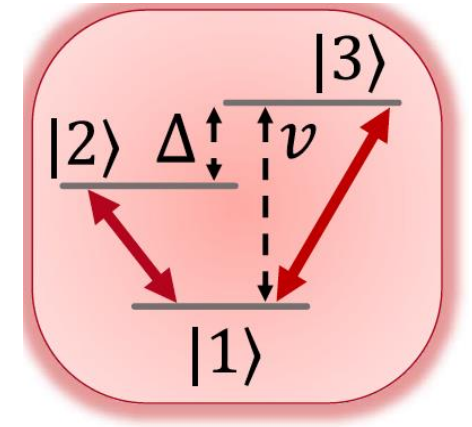
Counter intuitive!

$$\tau_1 \propto \left(\frac{\Omega}{\lambda} \right)^2 \frac{1}{T}$$

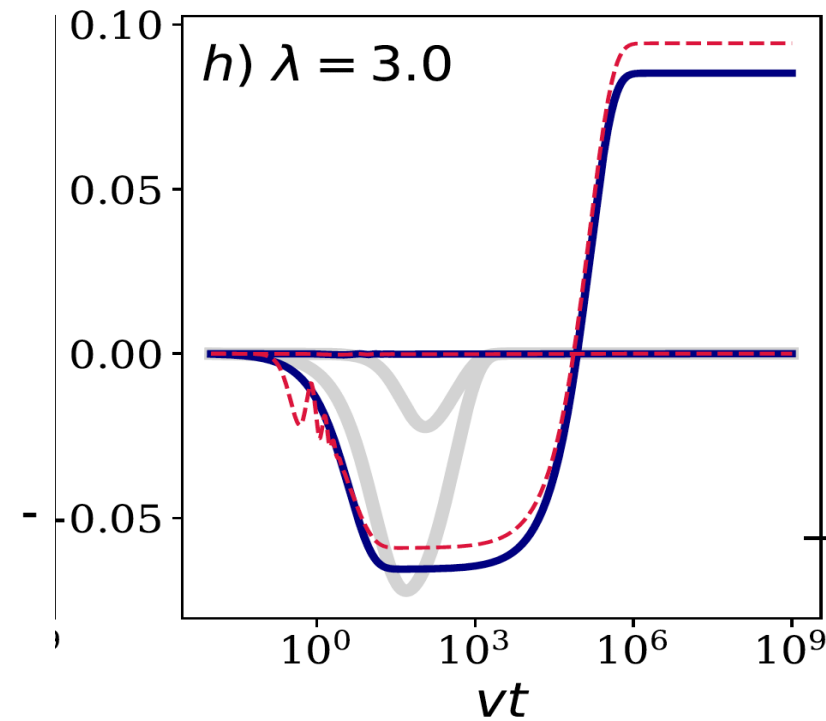
Intuitive

Separation of relaxation timescales via strong system–bath coupling: Dissipative three-level system as a case study

Brett Min, Matthew Gerry, Dvira Segal



Real part, coherence



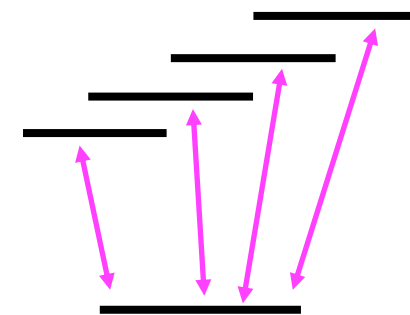
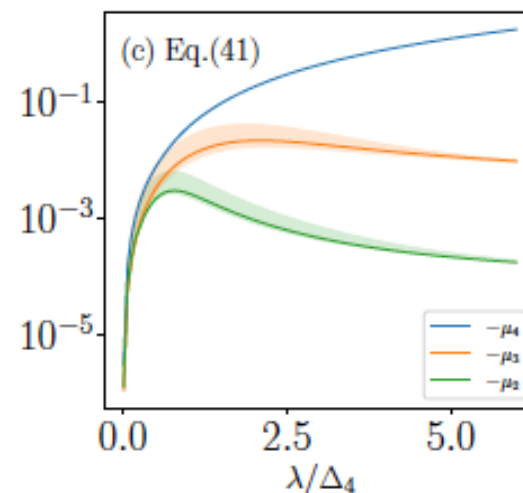
Hierarchical separation of relaxation timescales arrived via spectral localization bounds

Alex Stewart-Bozzo^{×,1,*} Jakub Garwoła^{×,1,†} Brett Min^{1,‡} and Dvira Segal^{2,1,§}

By deriving spectral localization bounds, we establish conditions under which **strong** system-bath coupling gives rise to a **hierarchy of relaxation timescales**. For the **generalized V model** family, we find that strong system–bath coupling gives rise to a **bright–dark** structure in the effective **system–bath coupling operator**: a single collective mode remains strongly coupled to the environment, while the remaining modes become progressively dark. **Numerical simulations** based on both secular and non-secular quantum master equations corroborate the emergence of timescale separation and the pronounced slowing down of dissipative dynamics at strong coupling. Our results reveal a **general mechanism** underlying **anomalously slow relaxation in strongly coupled open quantum systems** and provide a route for **engineering long-lived modes** through system–environment interactions.

$$\dot{\rho}_S = -i[H_S, \rho_S] + \sum_{j=1}^N \Gamma_j \left(L_j \rho_S L_j^\dagger - \frac{1}{2} \left\{ L_j^\dagger L_j, \rho_S \right\} \right)$$

$$\mu_k \in \left(-2\pi J_{\text{eff}}(\bar{E})[(k+1)\eta(\bar{E}) + 1] |p_k|^2, -2\pi J_{\text{eff}}(\bar{E})[\eta(\bar{E}) + 1] |p_k|^2 \right)$$



Outline: Today's talk

○ Introduction

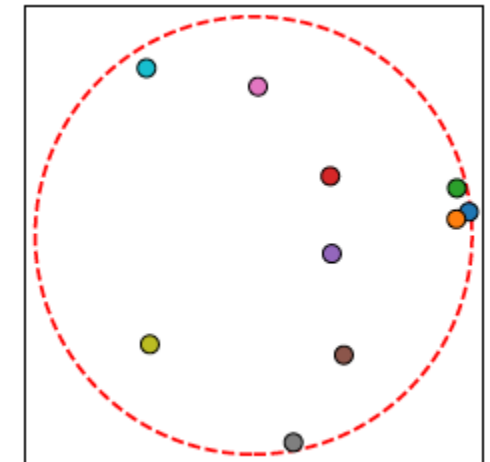
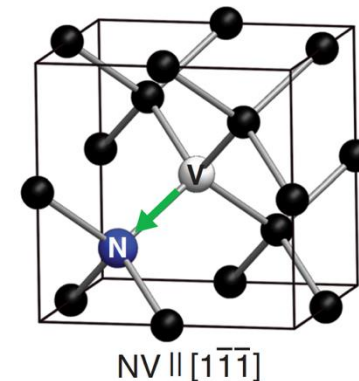
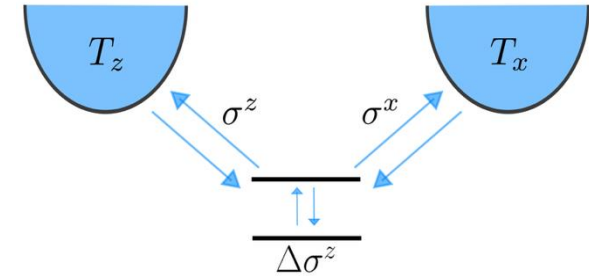
- T1 and T2 timescale and boring quantum dynamics

○ Strong coupling: -Nonadditive effects

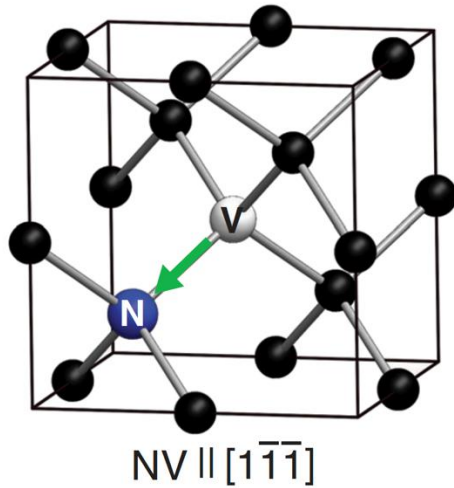
-Slow down - separation of relaxation timescale

○ Geometric heterogeneity: disorder with dephasing leading to long lived dynamics

○ Summary



Nitrogen-Vacancy (NV) diamond platform for quantum technologies



The qubit is encoded in the **electronic spin** of the defect.
The electronic ground state is a spin triplet with three levels $m_s=0, \pm 1$,
$$H = DS_z^2 + \gamma_e BS_z$$



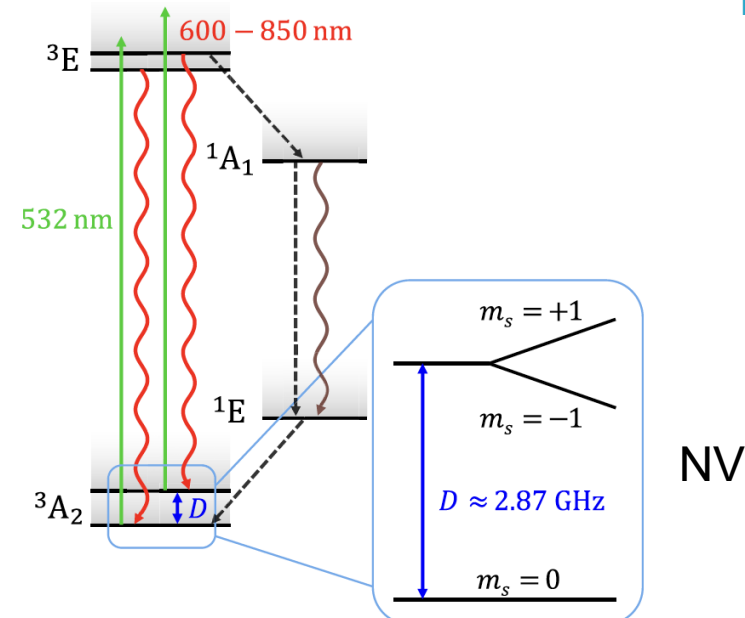
N. Bar-Gill (HUJI)

Advantages:

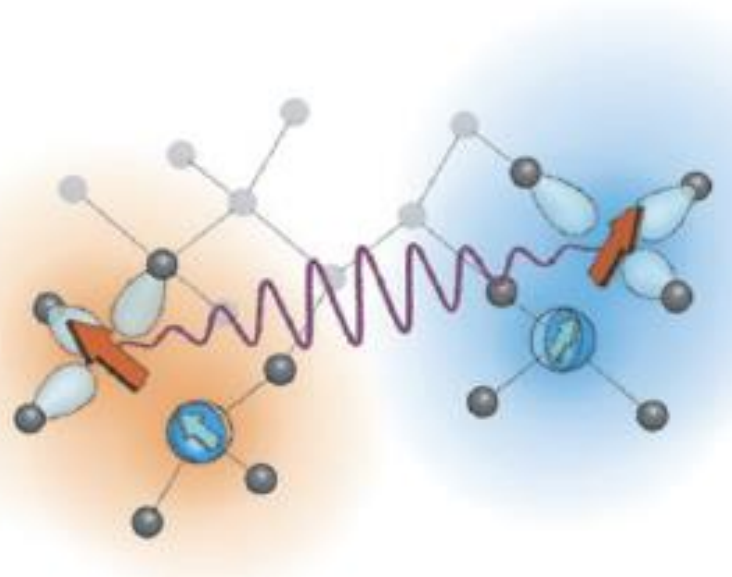
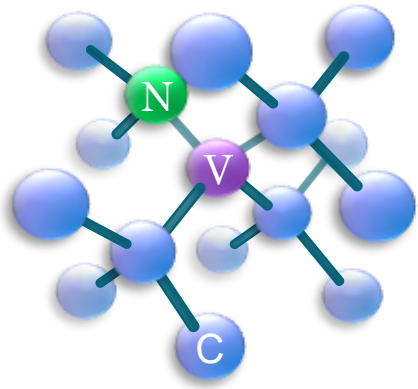
Solid-state system

Long coherence times

Optical addressing



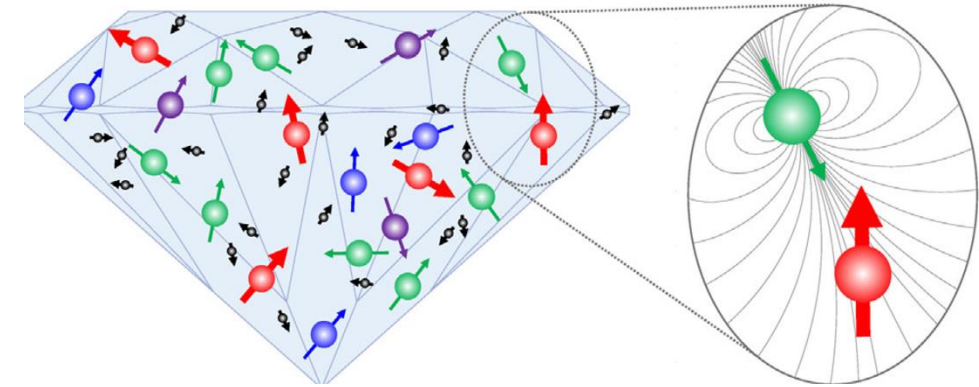
Nitrogen-Vacancy (NV) diamond platform for quantum technologies



Room-temperature entanglement between single defect spins in diamond

F. Dolde^{1*}, I. Jakobi¹, B. Naydenov^{1,2*}, N. Zhao¹, S. Pezzagna³, C. Trautmann^{4,5}, J. Meijer³, P. Neumann¹, F. Jelezko^{1,2} and J. Wrachtrup¹

Nature Physics 9, 139, (2013)



NV-centers (thick red arrows) experience magnetic fields caused by other spin defects in the diamond, including **substitutional nitrogen** (thin green arrows), ¹³C nuclei (small black arrows), and other **para-magnetic impurities** (blue and purple arrows).

Rev. Mod. Phys. **92**, 015004 (2020)

Quantum Transport in Disordered Spin Networks: Emergent Timescales and Competing Pathways

Roi Nevo,^{1,2} Brett Min,³ Maggie Lawrence,^{3,4} Dvira Segal,^{5,3,*} and Nir Bar-Gill^{2,1,†}

¹*The Racah Institute of Physics, The Hebrew University of Jerusalem, Jerusalem 91904, Israel*

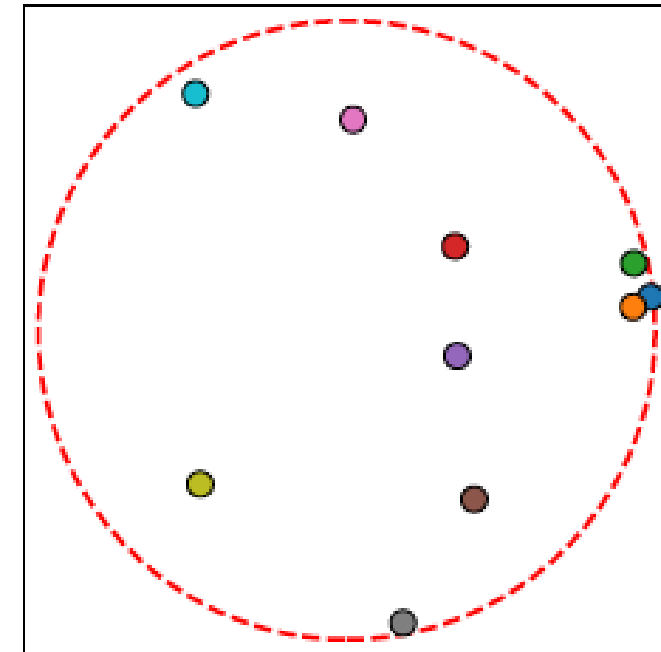
²*Inst. of Applied Physics, The Hebrew University of Jerusalem, Jerusalem 91904, Israel.*

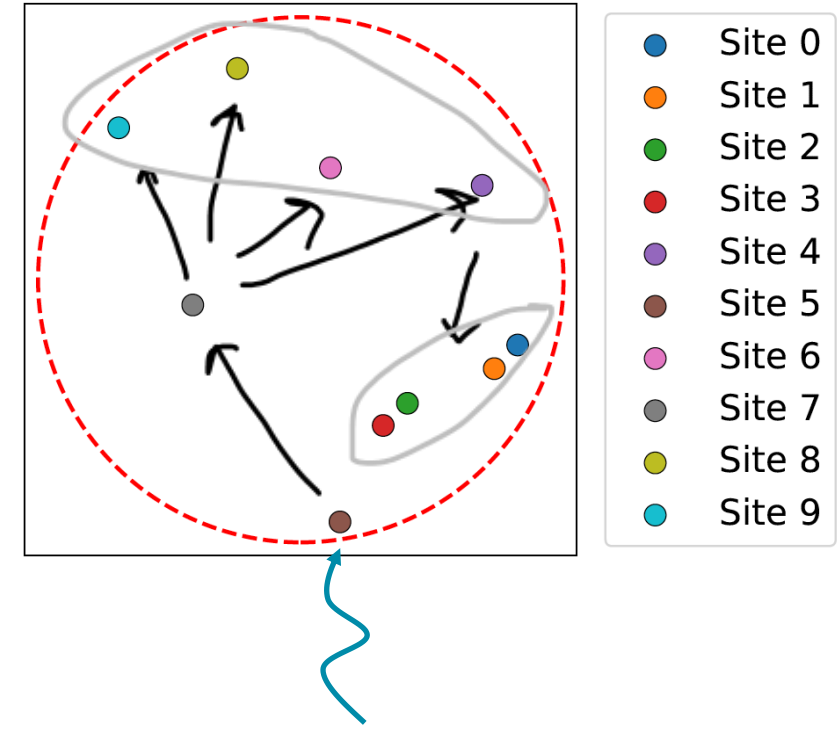
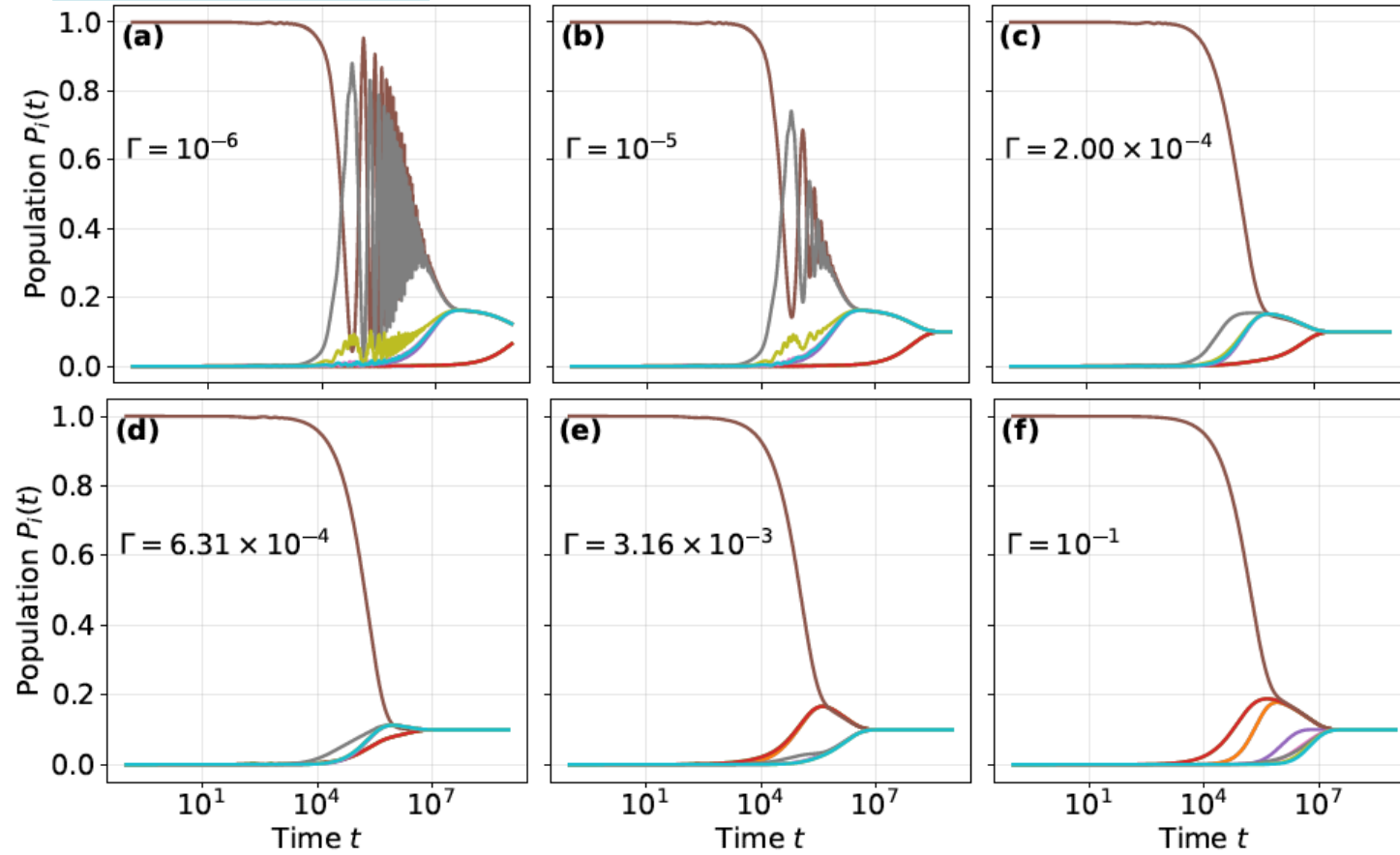
³*Department of Physics, 60 Saint George St., University of Toronto, Toronto, Ontario, M5S 1A7, Canada*

⁴*Vector Institute, W1140-108 College Street, Schwartz Reisman Innovation Campus Toronto, Ontario M5G 0C6, Canada*

⁵*Department of Chemistry and Centre for Quantum Information and Quantum Control, University of Toronto, 80 Saint George St., Toronto, Ontario, M5S 3H6, Canada*

We investigate the emergence of **multiple transport timescales** in the dissipative dynamics of a spin impurity coupled to a small, spatially **disordered** network of spins. Using a **two dimensional** tight-binding model with **dipolar interactions and local dephasing**, we demonstrate that **geometric heterogeneity leads to hierarchical coupling strengths**. This results in **pronounced separation of dynamical timescales**, far beyond those expected in homogeneous environments. By analyzing different metrics for dynamics, we identify distinct relaxation timescales associated with **cluster-level equilibration and global equilibration**.

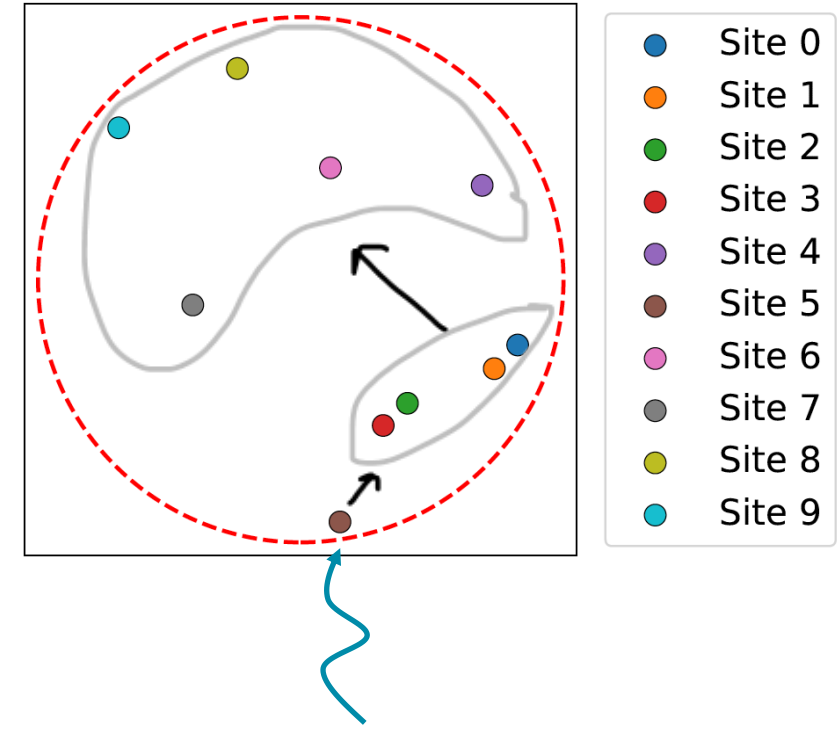
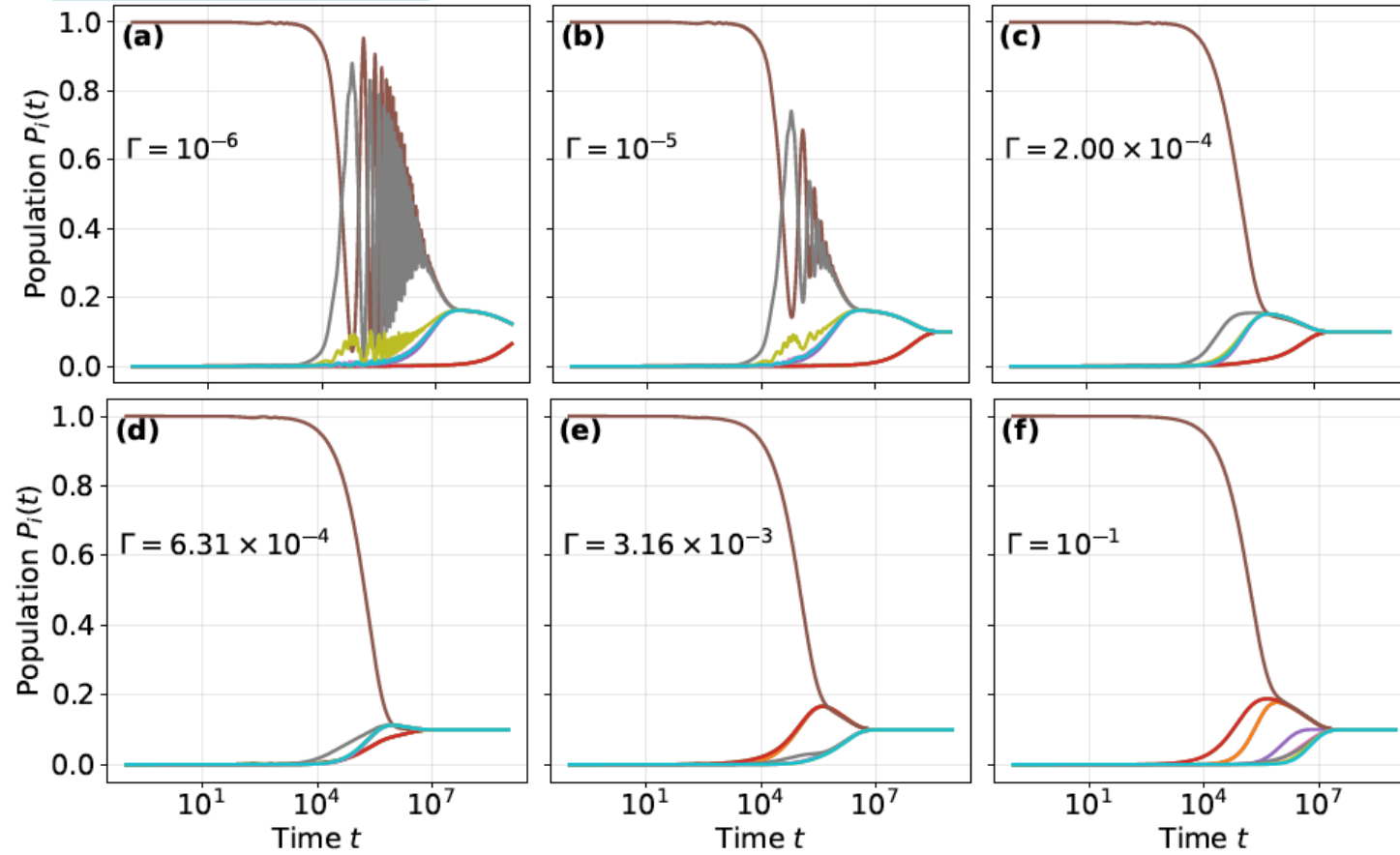




Competing pathways

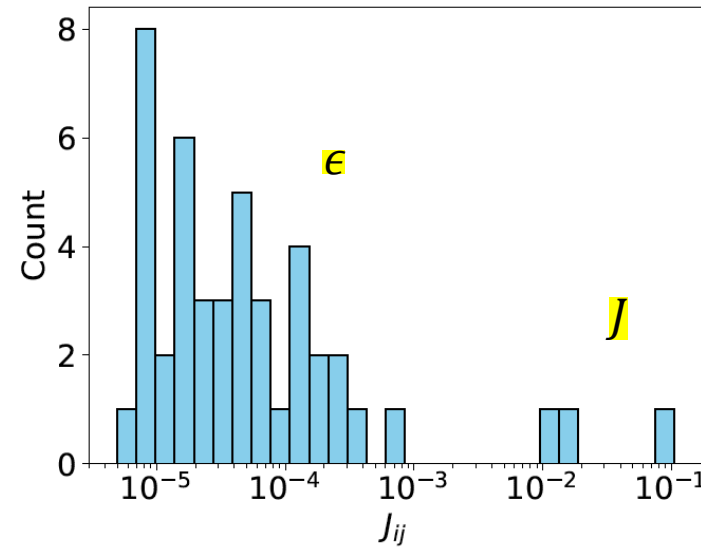
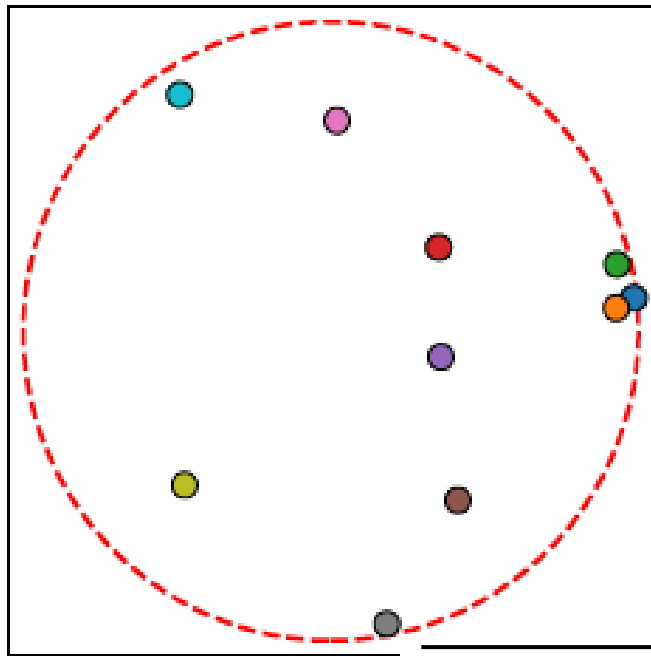
$$\hat{H}_S = \sum_{i=1}^N E_i |i\rangle \langle i| + \sum_{i \neq j} J_{ij} |i\rangle \langle j|$$

$$\dot{\rho}_S = -i[H_S, \rho_S] + \sum_{j=1}^N \Gamma_j \left(L_j \rho_S L_j^\dagger - \frac{1}{2} \{L_j^\dagger L_j, \rho_S\} \right)$$

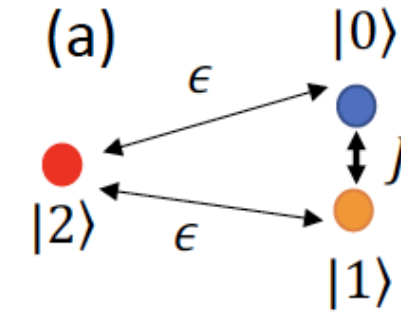


Competing pathways

$$\dot{\hat{\rho}} = -i \left[\hat{H}, \hat{\rho} \right] + \sum_{j=1}^N \Gamma_j \left(\hat{L}_j \hat{\rho} \hat{L}_j^\dagger - \frac{1}{2} \left\{ \hat{L}_j^\dagger \hat{L}_j, \hat{\rho} \right\} \right)$$



Slow Dynamics



[arXiv:2605.08918](https://arxiv.org/abs/2605.08918)

System / connectivity	Relaxation time	Numerical value
<i>(a) Weak dephasing regime $\Gamma \ll J$: $J = 10^{-2}$, $\epsilon = 10^{-4}$, $\Gamma = 10^{-5}$</i>		
Hybridized pair + remote site	$t_{\text{tr}} \sim \frac{J^2}{\epsilon^2 \Gamma}$	$\sim 10^9$
Homogeneous baths (coupling J)	$t_{\text{tr}} \sim \frac{1}{\Gamma}$	$\sim 10^5$
Homogeneous bath (coupling ϵ)	$t_{\text{tr}} \sim \frac{1}{\Gamma}$	$\sim 10^5$

Summary: Anomalous Dynamics

- **Strong system-bath coupling**

→ nonadditivity, frustration, polaron undressing

→ Bright-dark states: separation of timescales

- **Network heterogeneity (disorder) and dephasing**

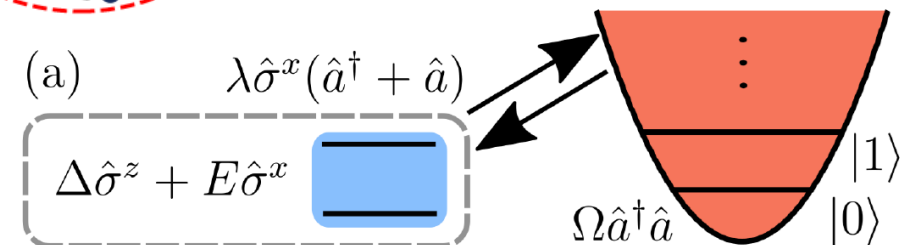
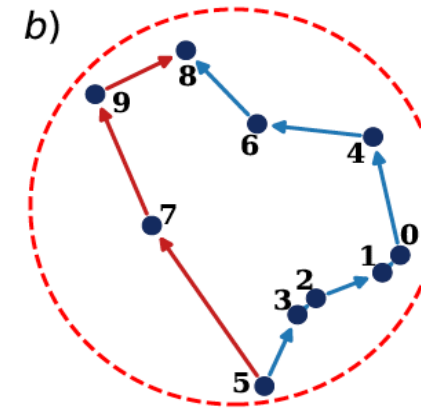
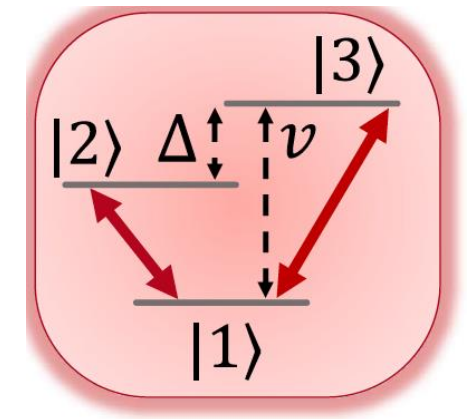
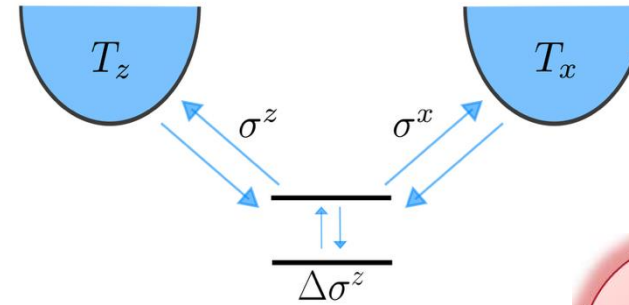
→ Long relaxation timescales

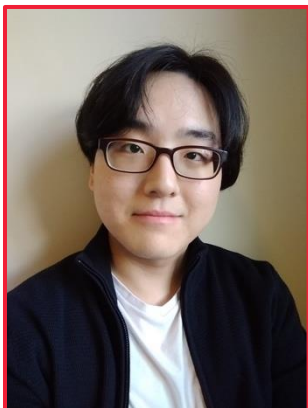
→ Competing pathways, multiple transport optima

- **Quasidegeneracy in the spectrum**

→ Long lived coherences

→ Strong Mpemba effect

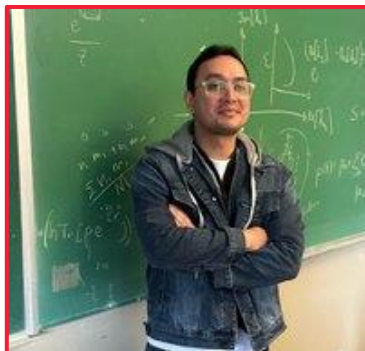




Brett Min
(GS)



Nicholas
Anto-Sztrikacs
(GS→Scotia Bank)



Marlon Brenes
(PDF→Costa Rica)



Jakub Garwola
(GS)



Jitian Chen
(UG→med school)



Maggie Lawrence (Grad)



Alessandro
Prositto (Grad)



Madeline Forbes
(UG UBC)



Matthew Pocrnic (GS)



Prof. Nir Bar-Gill
(HUJI)



Roi Nevo
(grad student)